
机器人学

第四章：运动学

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-
- 机械臂正运动学
 - 机械臂逆运动学

引言 -1

□ 运动学(Kinematics): 讨论运动状态本身, 未涉及到产生运动的「力」

◆ 位置(x)、速度(v)、加速度(a)、和时间(t) 之间的关系

◆ 移动/转动

◦ 位置/姿态

◦ 速度/角速度

◦ 加速度/角加速度

$$\begin{aligned} v &= \frac{d}{d} x & a &= \frac{d}{d} v \\ a &= \frac{d^2}{dt^2} x & v dv &= a ds \end{aligned}$$

□ 动力学(Dynamics): 讨论力/力矩如何产生运动

◆ Newton's 2nd Law

$$\sum F = ma$$

◆ Work & energy

$$T_1 + V_1 + U_{1-2}' = T_2 + V_2$$

◆ Impulse & momentum

$$\int \sum F dt = G_2 - G_1$$

引言 -2

□ 机械手臂

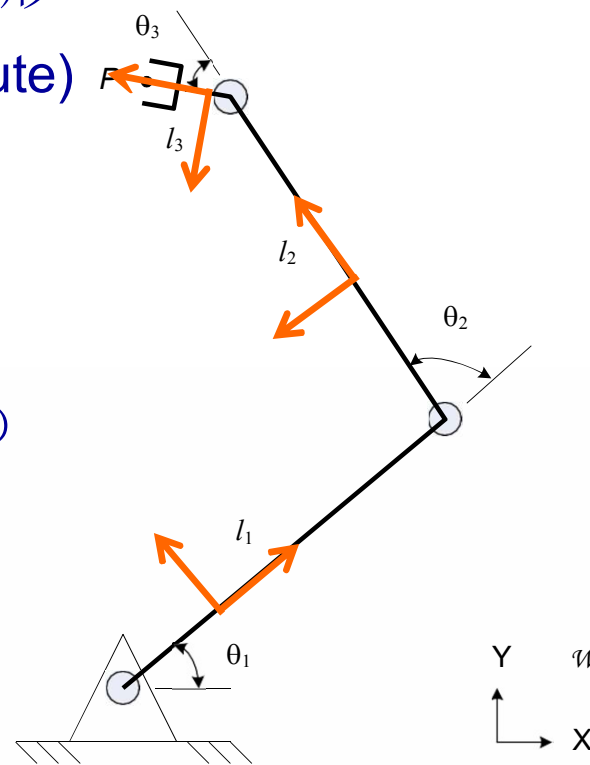
- ◆ 多个连杆(连杆)相串连，具有复杂的几何外形
- ◆ 连杆间可相对 移动(prismatic)或转动(revolute)
由致动器驱动来达成

□ 对应关系

- ◆ 需求：手臂末端点状态（位置 $^W P$ 、速度...）
- ◆ 达成方式：驱动各致动器
$$^W P = f(\theta_1, \theta_2, \dots, \theta_n)$$

□ 描述手臂状态方法

- ◆ 找出连杆间的相对几何状态
- ◆ 在各连杆上建立frame，以frame状态来代表连杆状态



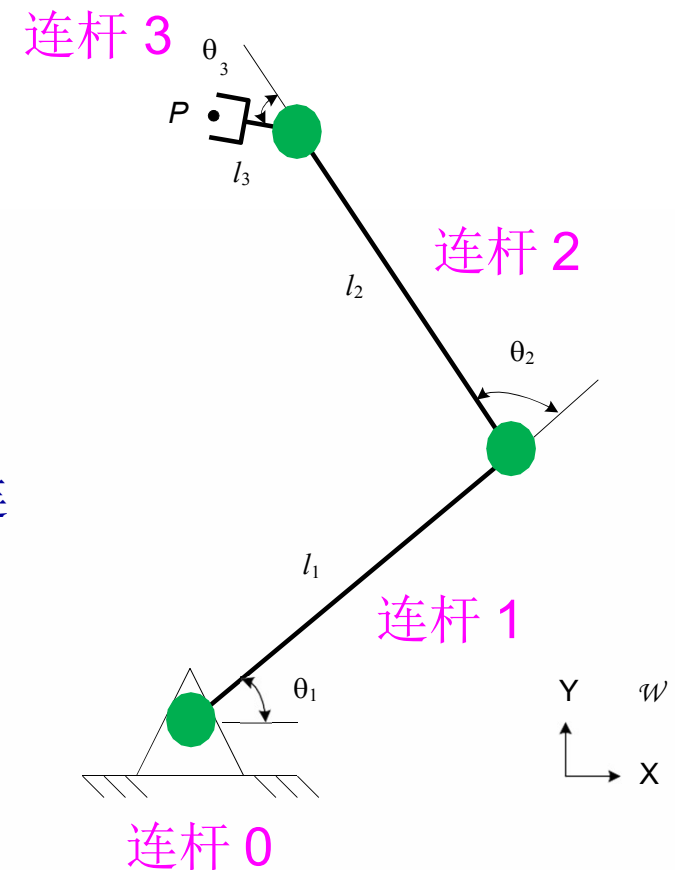
手臂几何描述方式 -1

□ 关节

- ◆ 每个旋转或移动的关节具有 1 DOF
- ◆ 每个关节对 某特定轴 进行旋转或者移动

□ 连杆

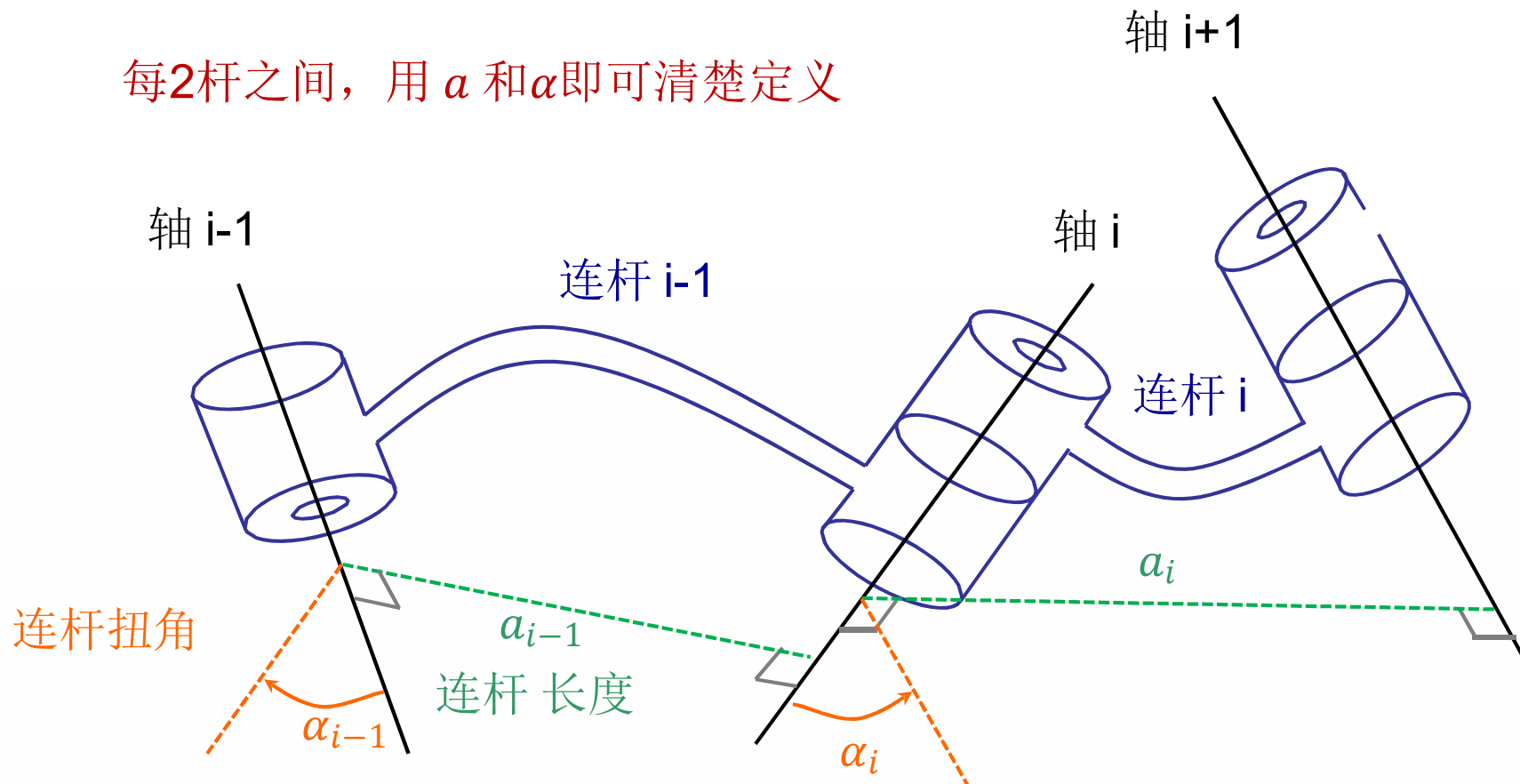
- ◆ 连接关节的连杆，为刚体(rigid body)
- ◆ 编号方式
 - 连杆 0: 基座 (base)，不动的连杆
 - 连杆 1: 和连杆 0 相连，第一个可动的连杆
 - 连杆 2: 第二个可动的连杆
 - 依序下去...



手臂几何描述方式 -2

- 对空间中2个任意方向的轴，两轴之间具有一线段和此2个轴都相互垂直

每2杆之间，用 a 和 α 即可清楚定义



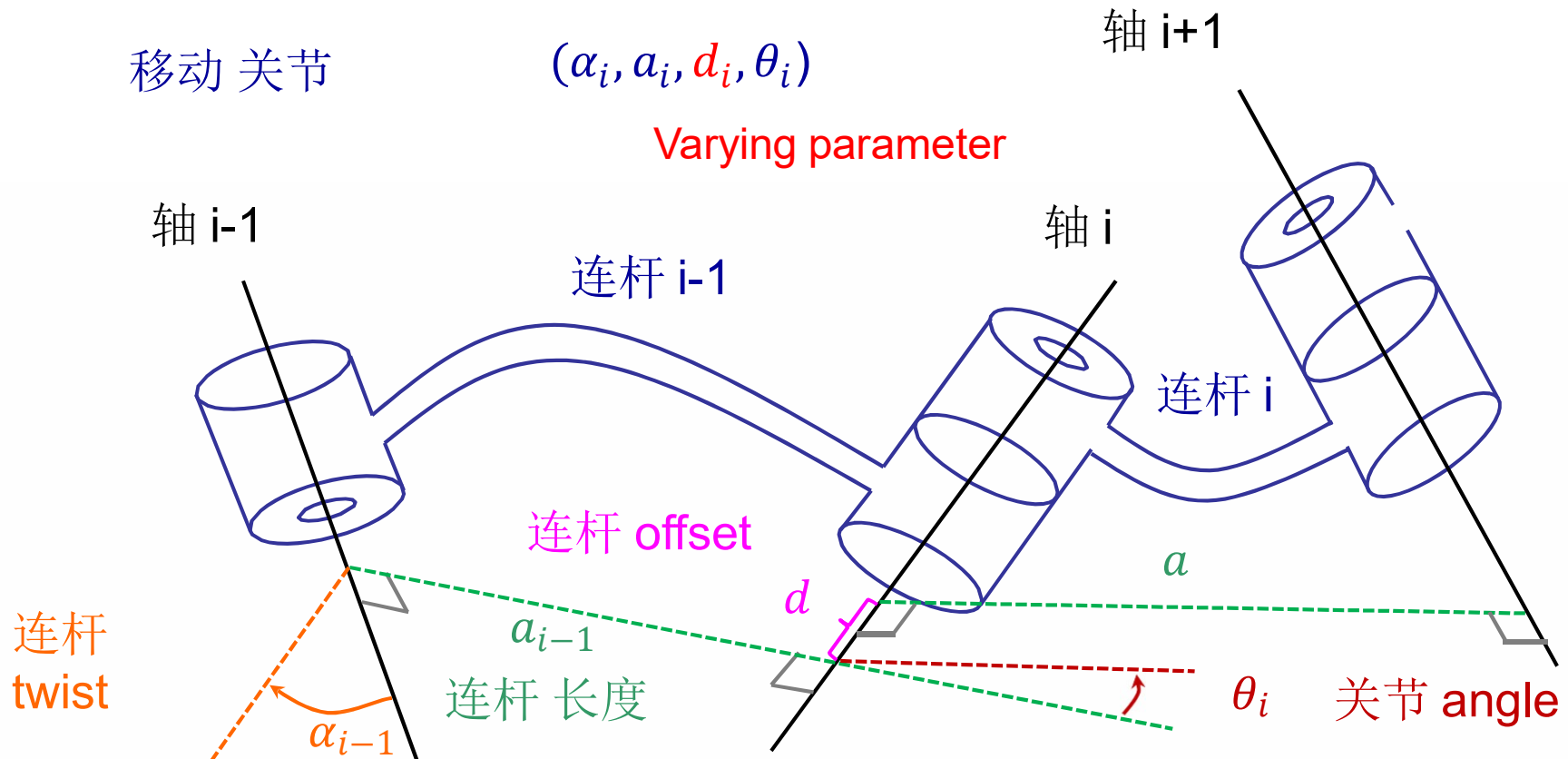
手臂几何描述方式 -3

- 但若要多杆串连, 则另需要两个参数(d_i, θ_i), 来描述相邻 线段 a_{i-1} 和 a_i 间的相对几何关系

旋转 关节 $(\alpha_i, a_i, d_i, \theta_i)$

移动 关节 $(\alpha_i, a_i, d_i, \theta_i)$

Varying parameter



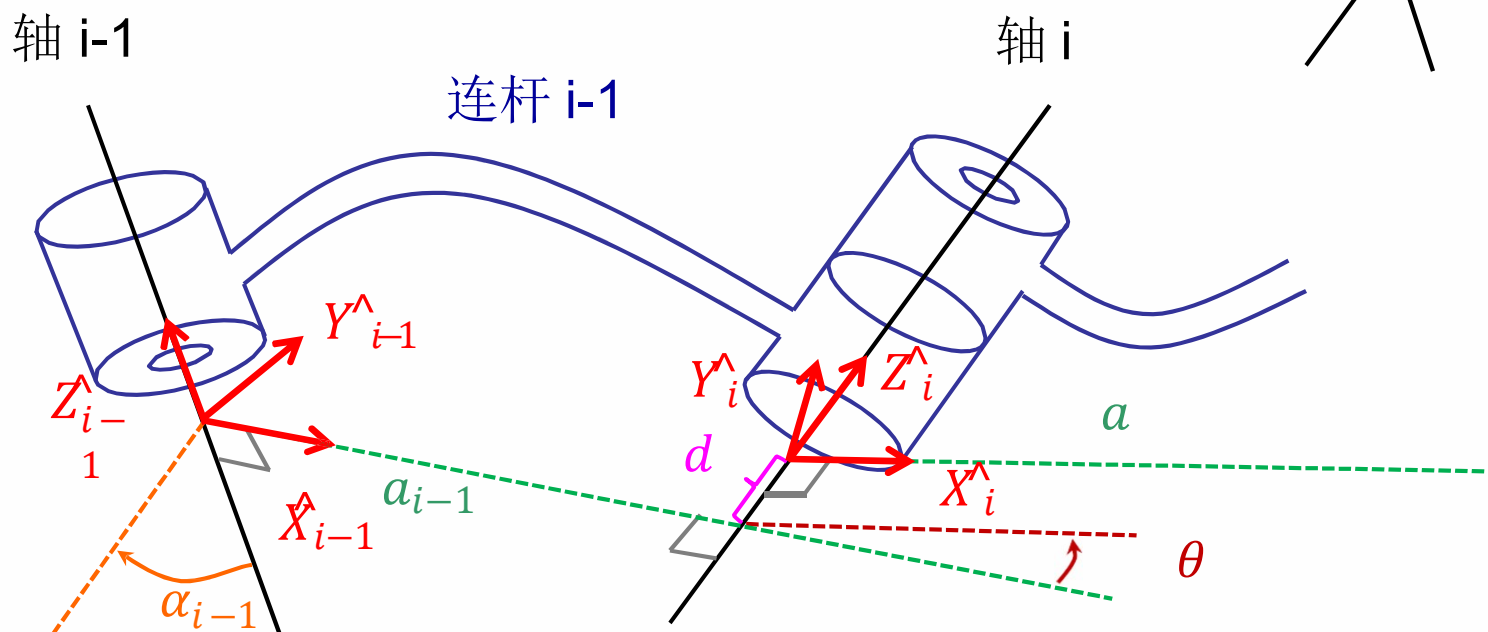
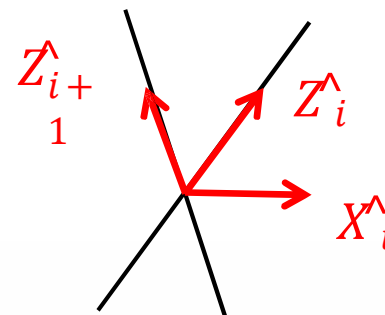
连杆上建立坐标系 -1

□ Z_i 转动或移动轴的方向

□ X_i 沿着 a_i 方向(if $a_i \neq 0$)

和 Z_i 和 Z_{i+1} 两者垂直 (if $a_i = 0$)

□ Y_i 与 X_i 和 Z_i 两者垂直, 依循右手定则



连杆上建立坐标系 -2

□ 基座 连杆 (0)

Frame {0} 与 frame {1} 重合

$$a_0 = 0$$

$$\alpha_0 = 0$$

旋转关节

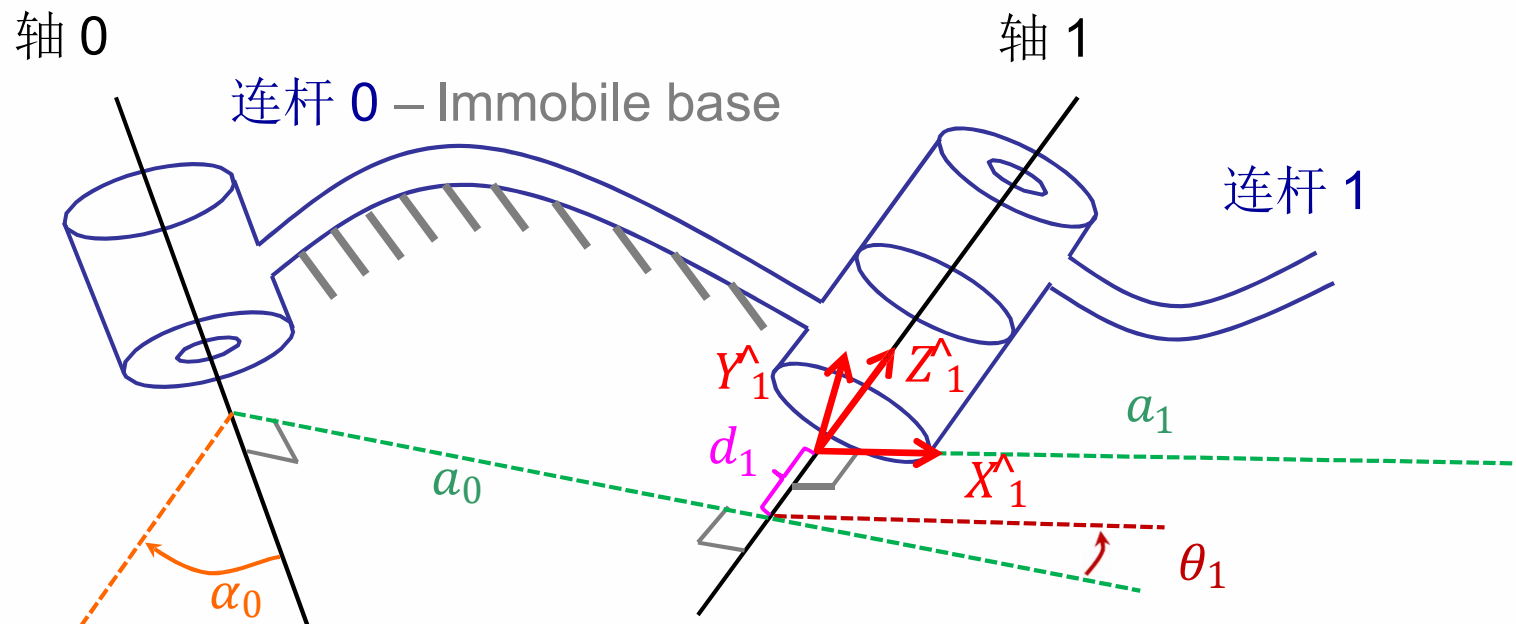
θ_1 任意

$$d_1 = 0$$

平移关节

d_1 任意

$$\theta_1 = 0$$



连杆上建立坐标系 -3

□ Last 连杆 (n)

取和 $\hat{X}_{n-1}^{\wedge}$ 同方向

$$a_n = 0 \quad \alpha_n = 0$$

旋转 关节

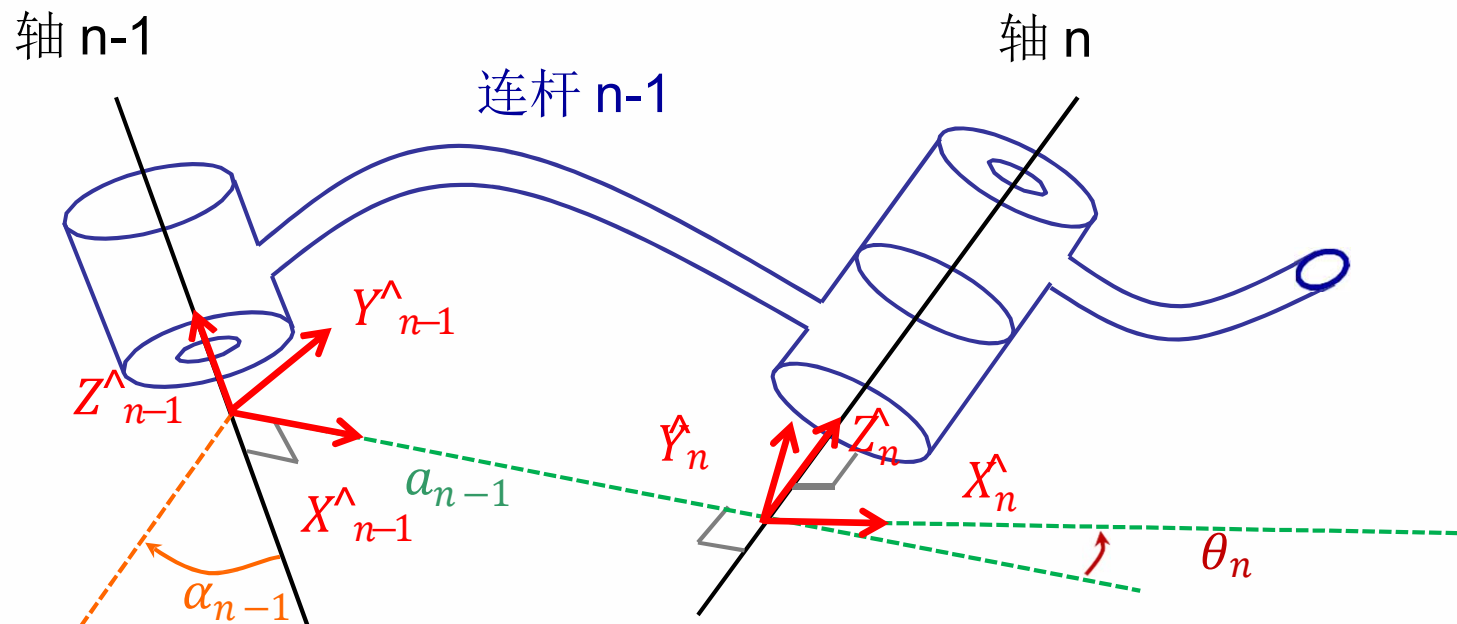
θ_n 变量

$$d_n = 0$$

平移 关节

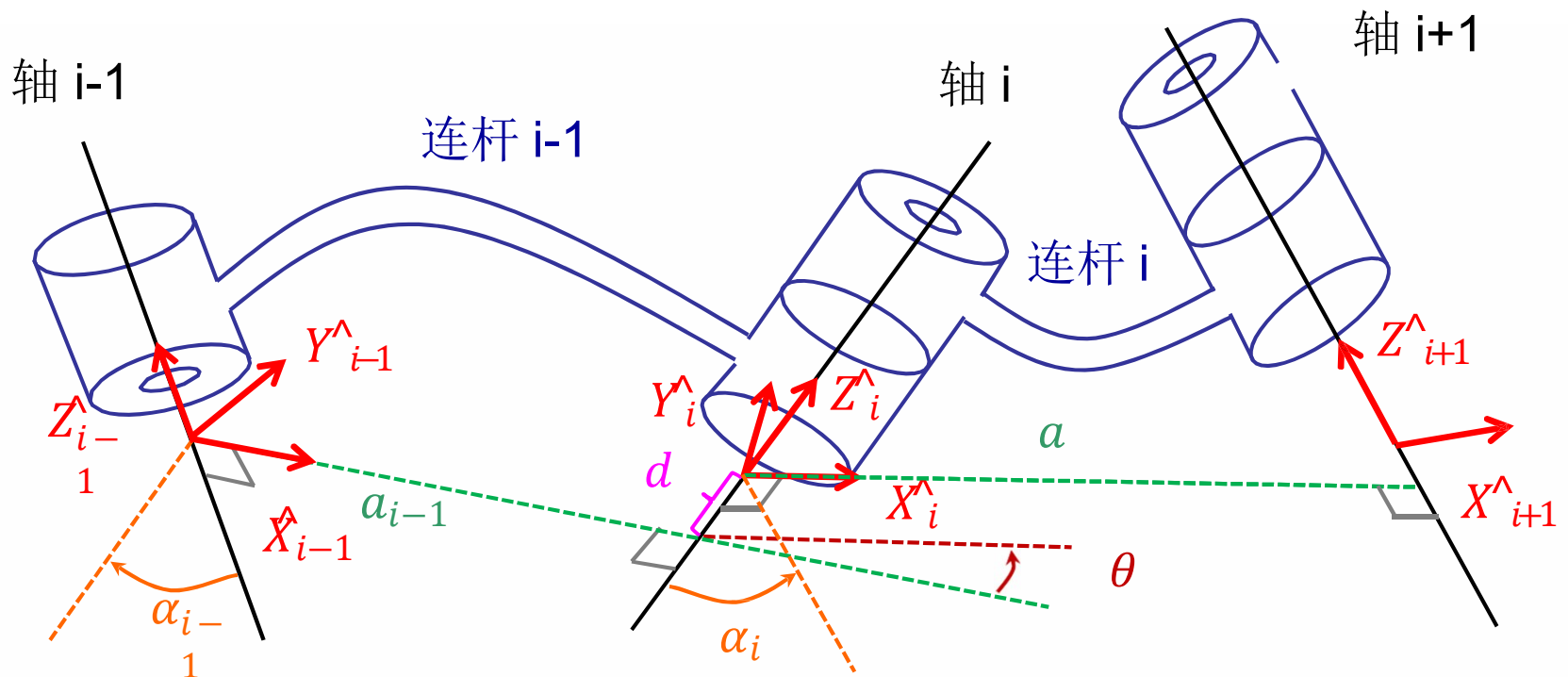
d_n 变量

$$\theta_n = 0$$



Denavit-Hartenberg表达法 (Craig version)

- α_{i-1} : 以 X_{i-1} 方向看, Z_{i-1} 和 Z_i 间的夹角
- a_{i-1} : 沿着 X_{i-1} 方向, Z_{i-1} 和 Z_i 间的距离 ($a_i > 0$)
- θ_i : 以 Z_i 方向看, X_{i-1} 和 X_i 间的夹角
- d_i : 沿着 Z_i 方向, X_{i-1} 和 X_i 间的距离

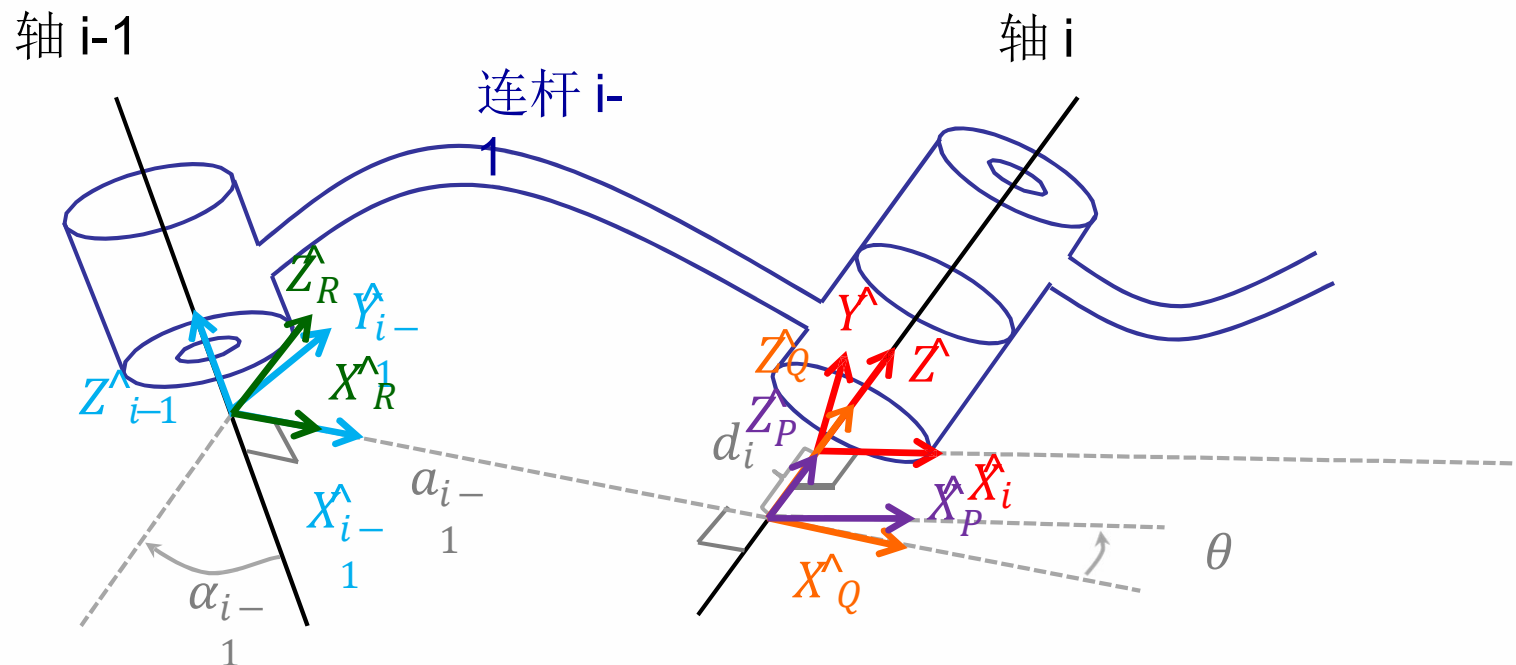


连杆 Transformations -1

$${}^{i-1}P = {}^{i-1}T {}^iP$$

$${}^{i-1}P = {}^{i-1}T_R T_Q^R T_P^Q T^P {}^iP$$

$$\begin{aligned} {}^{i-1}T &= {}^{i-1}T_R T_Q^R T_P^Q T^P {}^i \\ &= T_{X_{i-1}}(\alpha_{i-1}) T_{X_R}(a_{i-1}) T_{Z_Q}(\theta_i) T_{Z_P}(d_i) \end{aligned}$$



连杆 Transformations -2

□ 于是有

$$\begin{aligned} {}^{i-1}T &= T_{X_{i-1}}^{\wedge}(\alpha_{i-1})T_{X_R}^{\wedge}(a_{i-1})T_{Z_Q}^{\wedge}(\theta_i)T_{Z_P}^{\wedge}(d_i) \\ &= \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

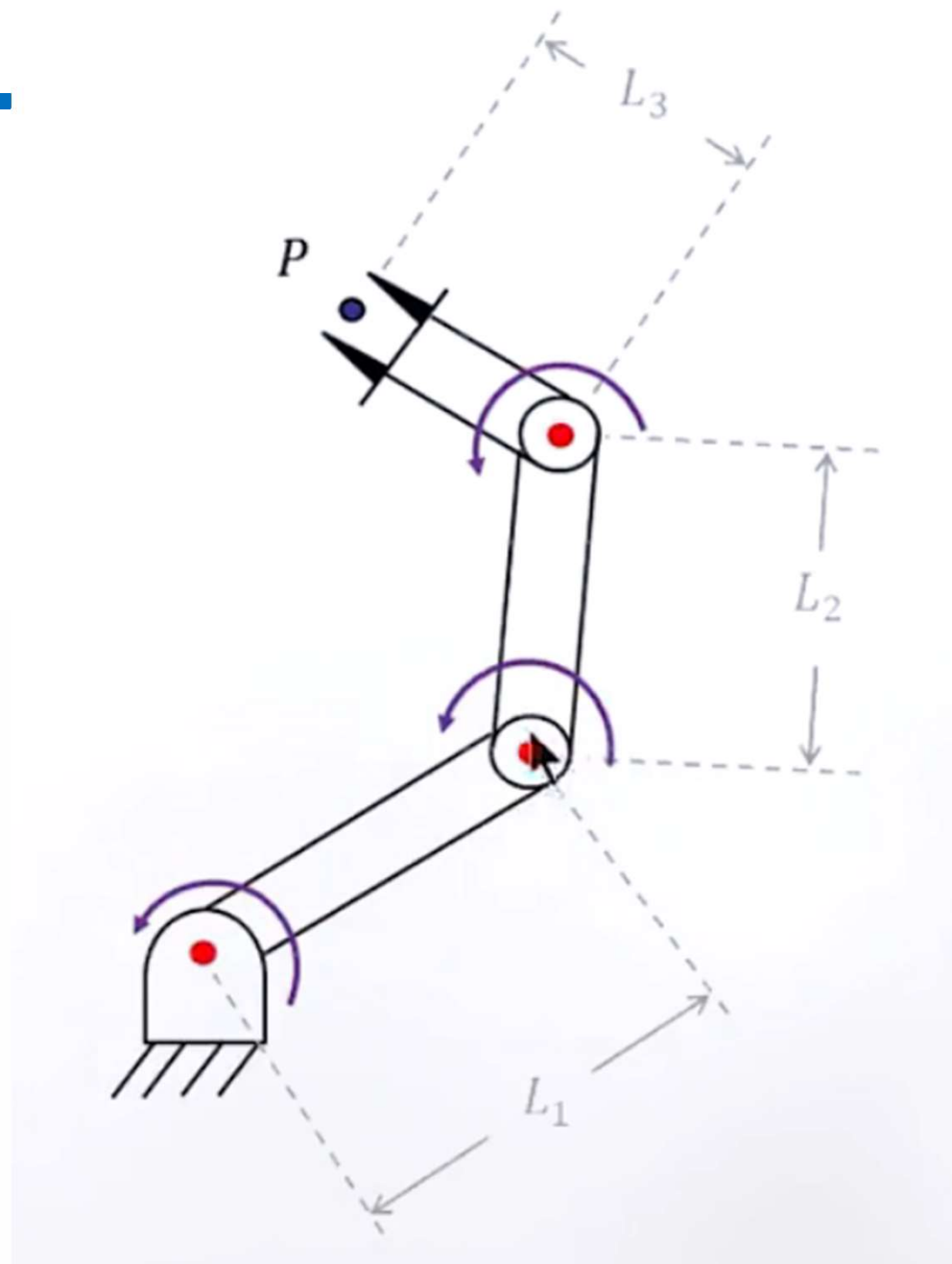
□ 连续连杆 transformations

$${}^0T_n = {}^0T_1 {}^1T_2 {}^2T_3 \dots {}^{n-2}T_{n-1} {}^{n-1}T_n$$

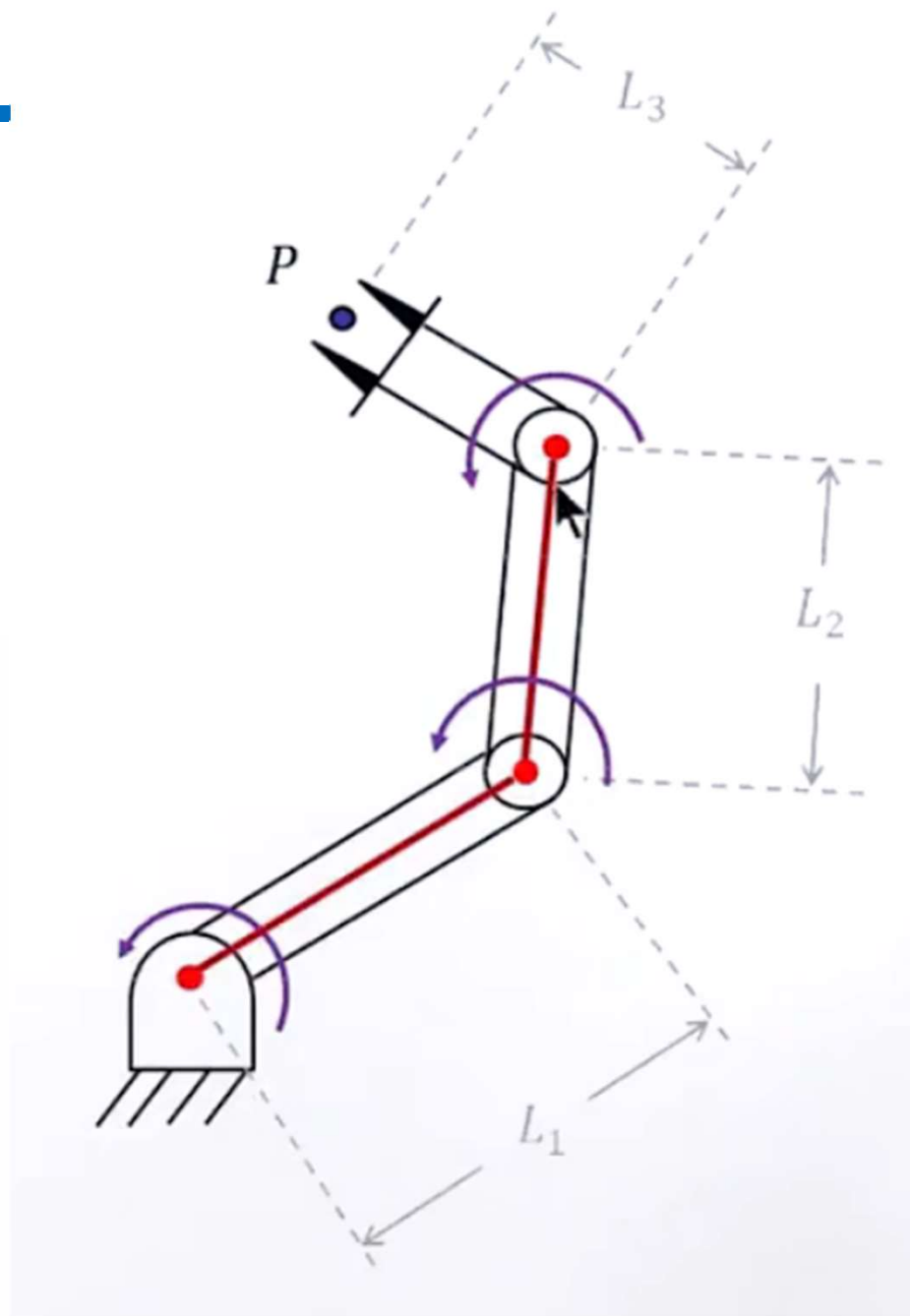
Frame {n} 相对于 Frame {0} 的空间几何关系具清楚且量化之定义

在Frame {n} 下表达的向量可转回 Frame {0} 下来表达

关节轴

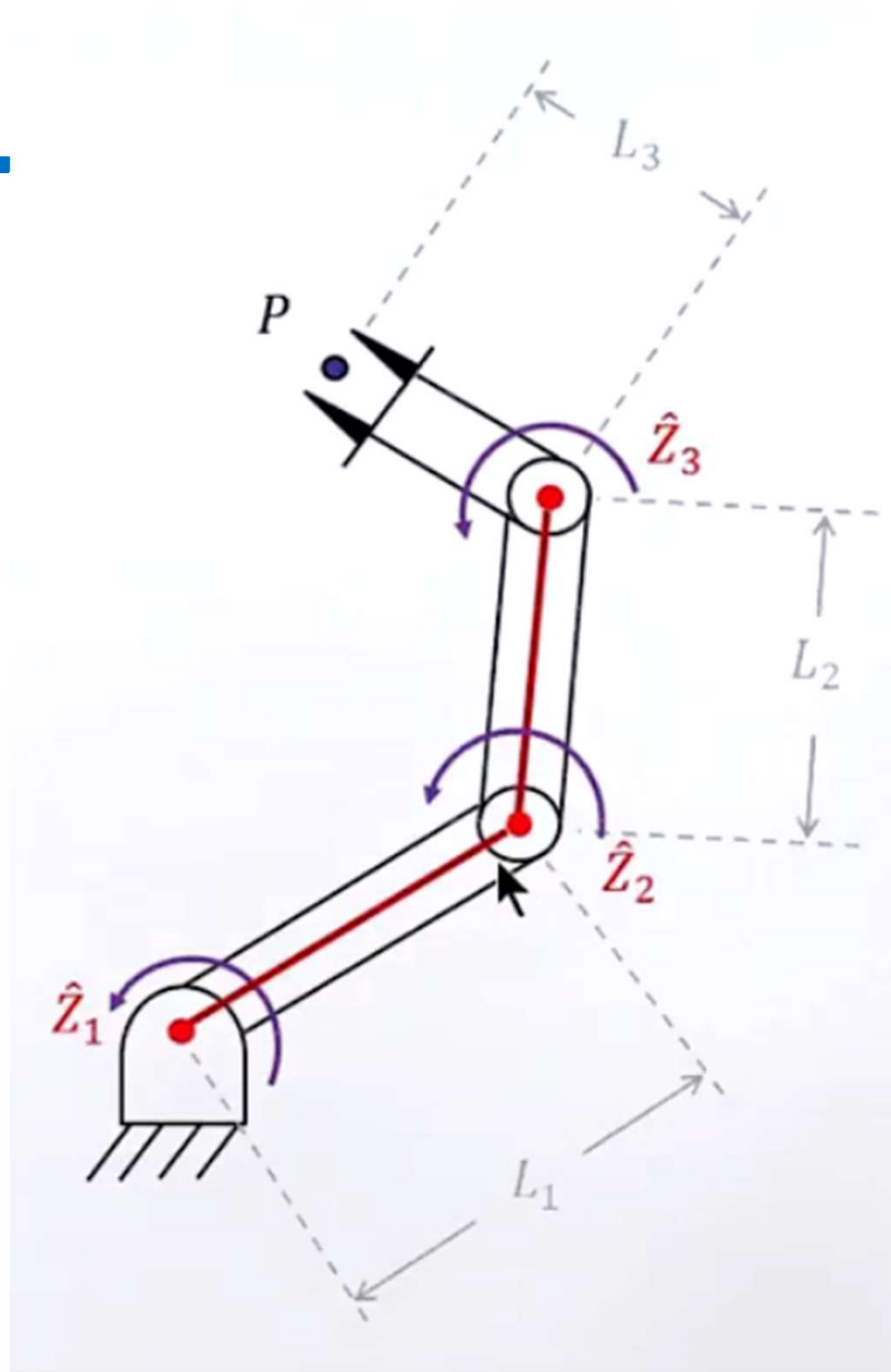


关节轴 公垂线



关节轴
公垂线

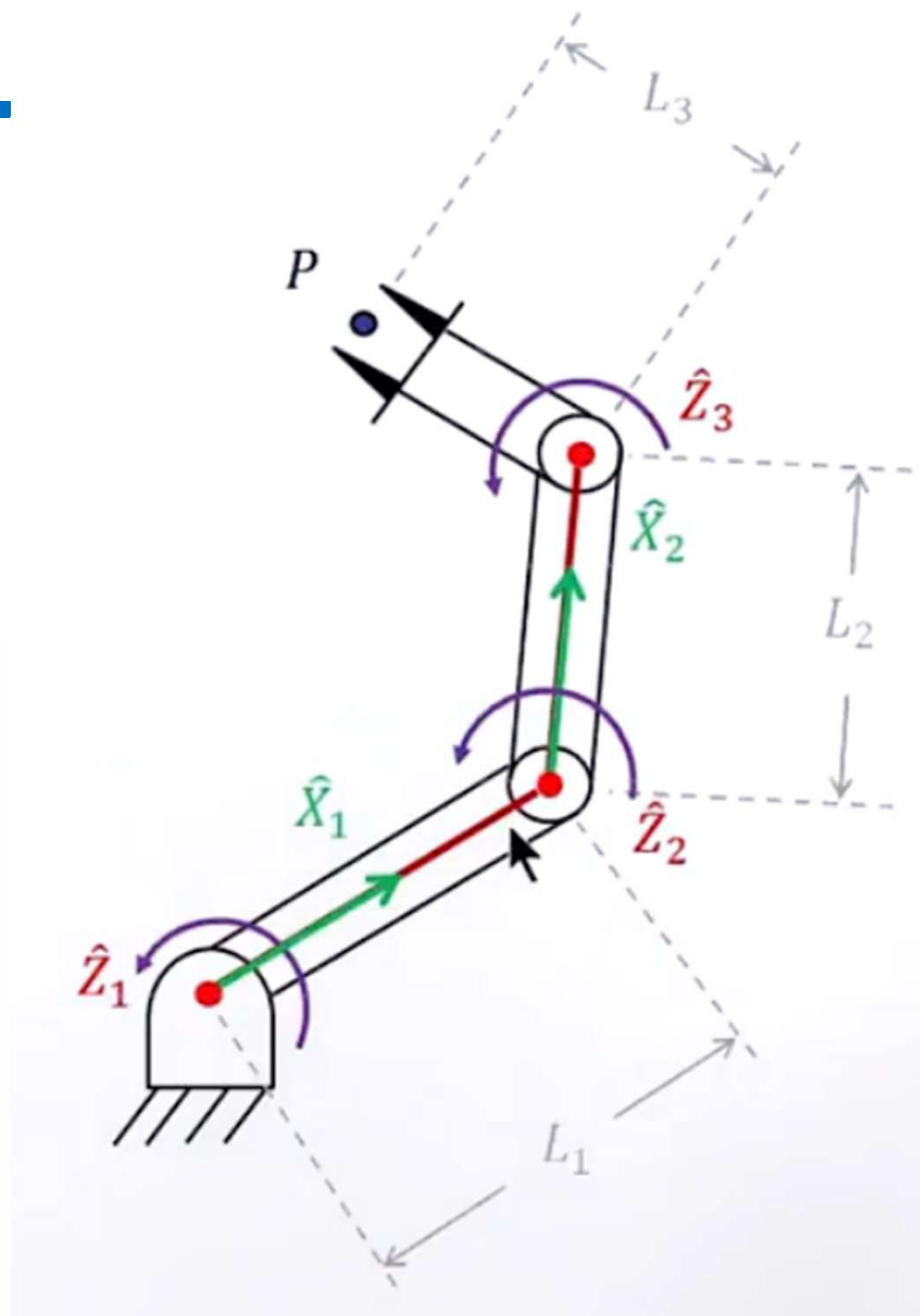
Z_i



关节轴
公垂线

Z_i

$\hat{X}_i$

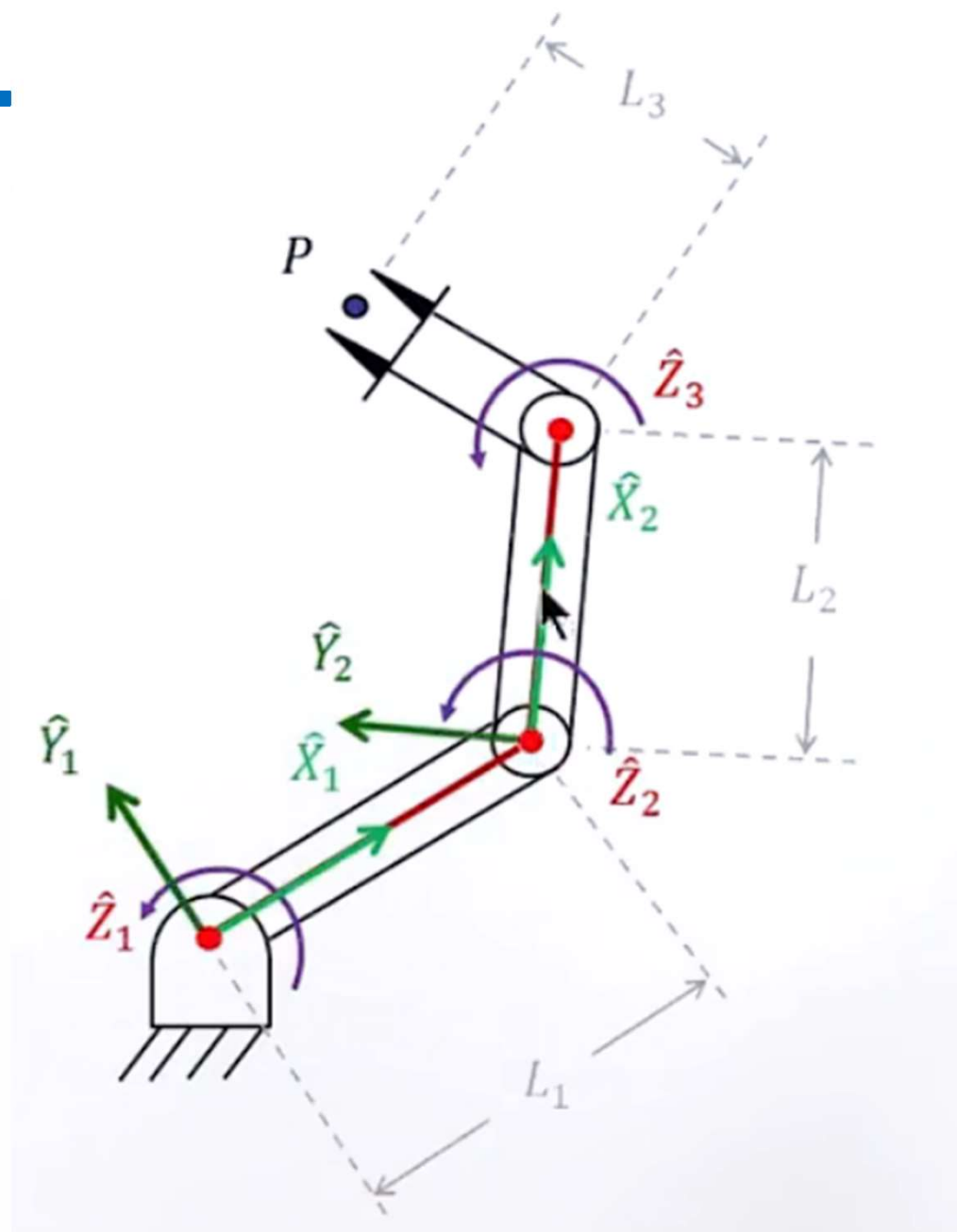


关节轴
公垂线

Z_i

$\hat{X}_i$

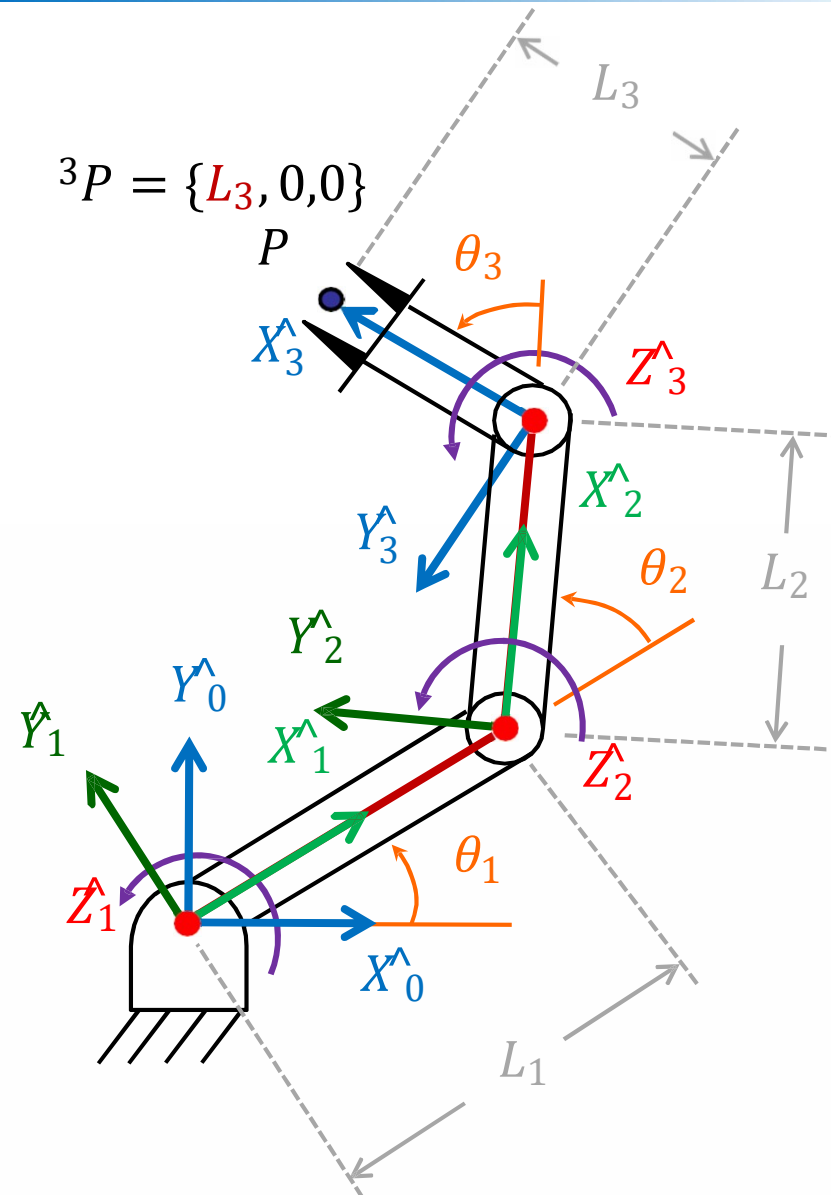
$\hat{Y}__i$



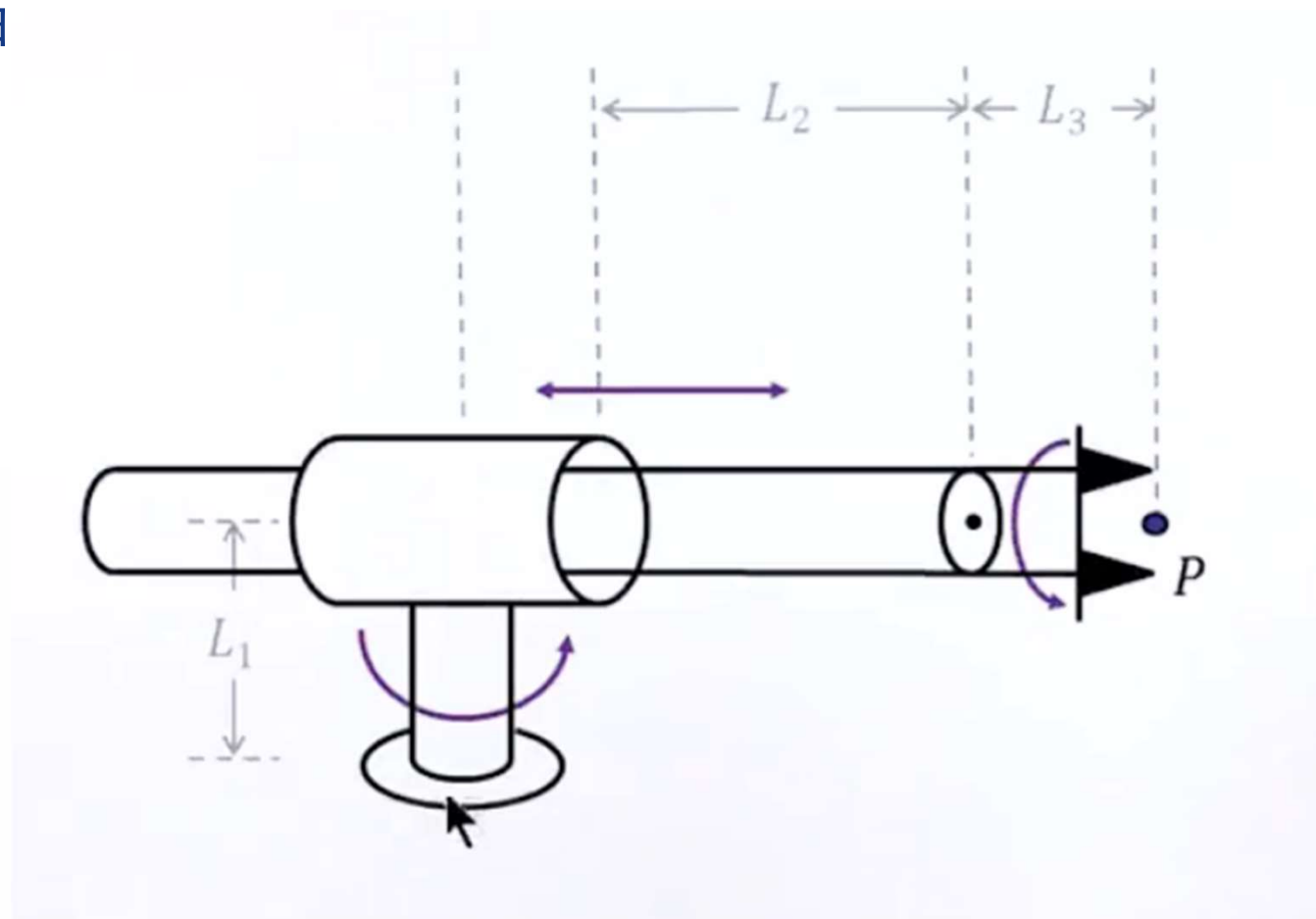
Example: A RRR Manipulator

- 关节 轴
- Common perpendiculars
- $\hat{Z}_i$
- $\hat{X}_i$
- $\hat{Y}_i$
- 坐标系 $\{0\}$ and $\{n\}$

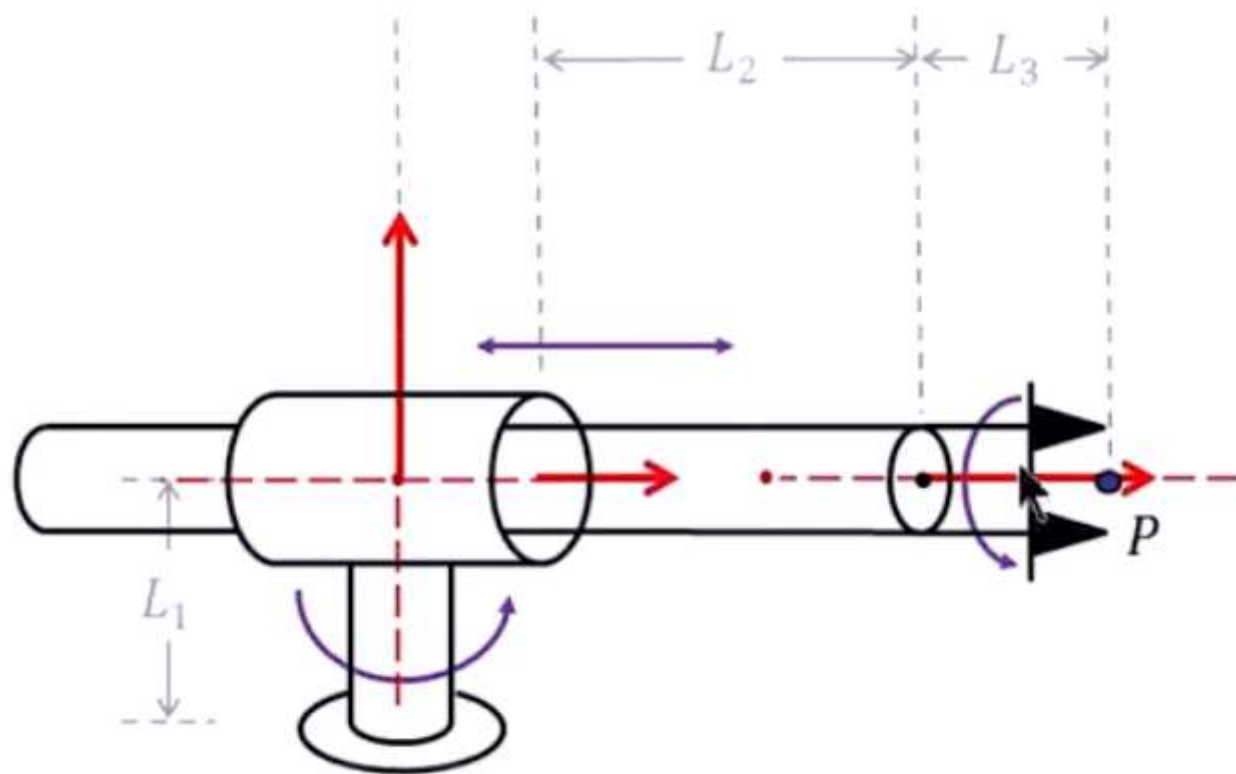
i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3



关节轴



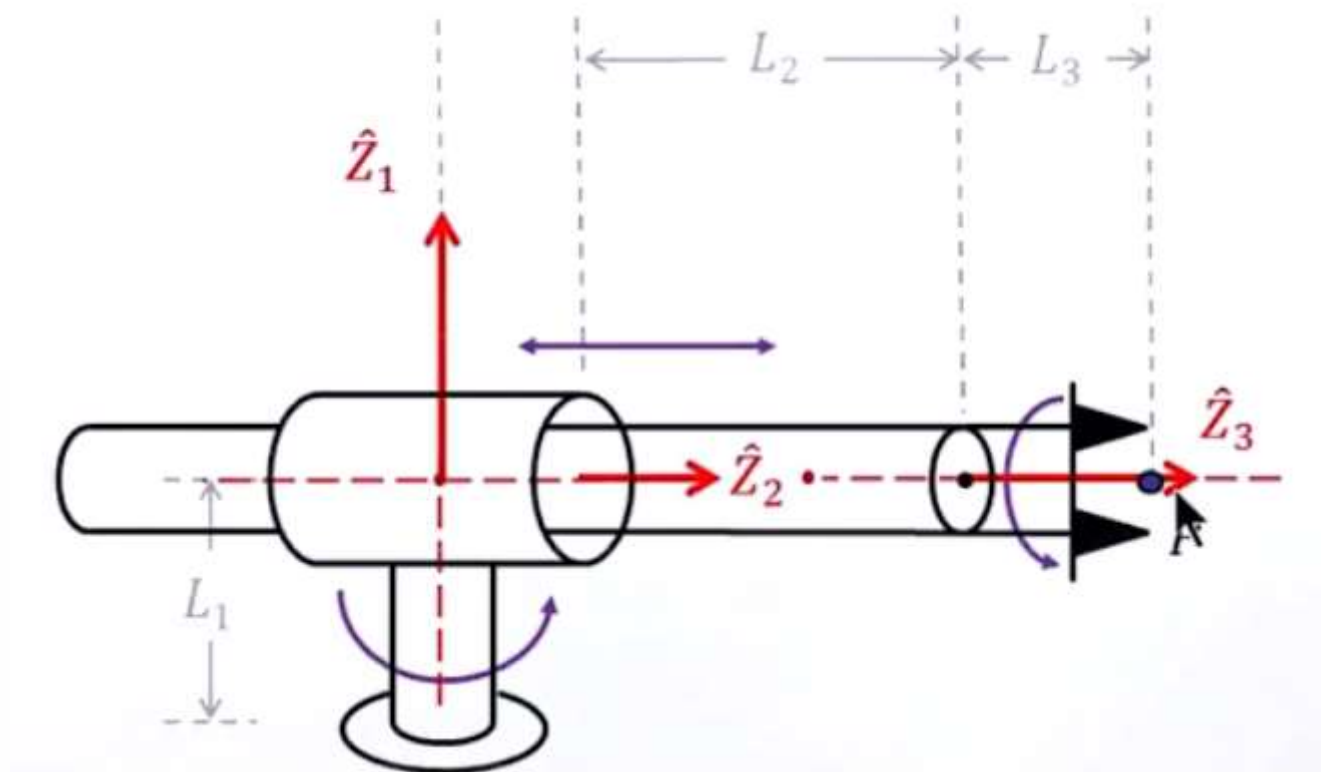
关节轴 公垂线



关节轴

公垂线

$\hat{Z}_i$

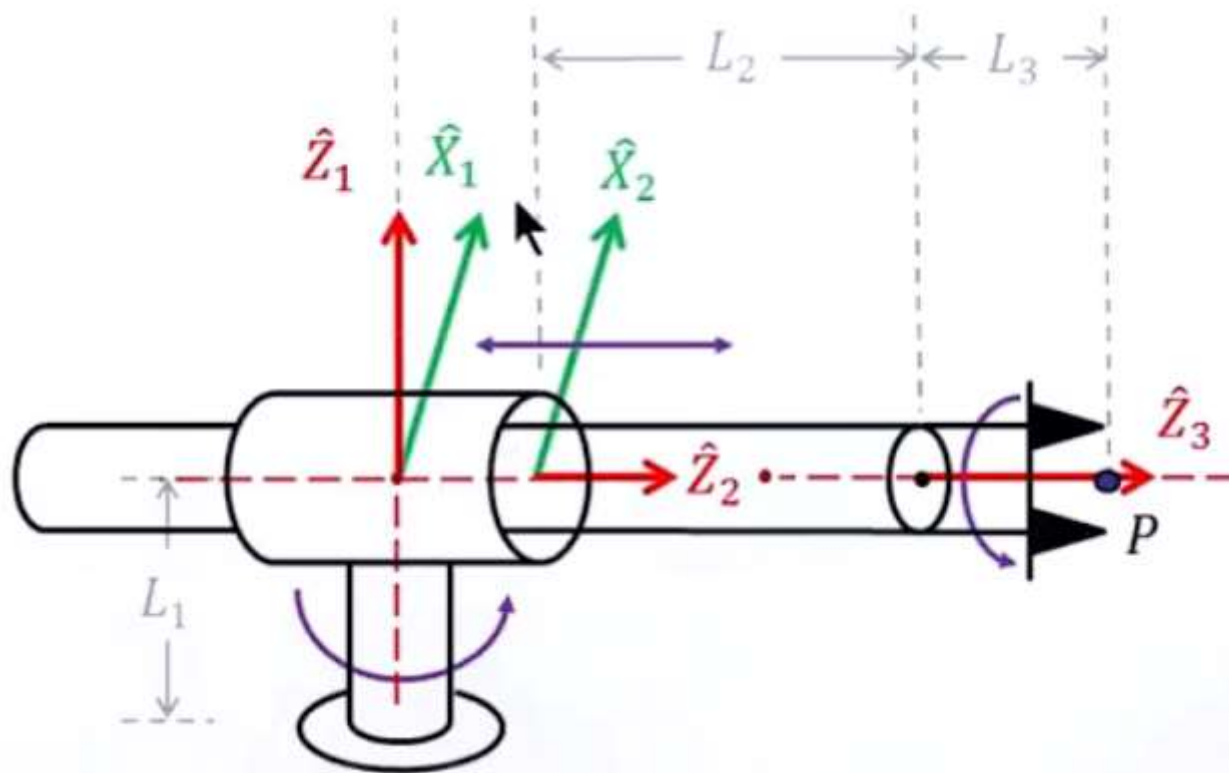


关节轴

公垂线

$\hat{Z}_i$

$\hat{X}_i$



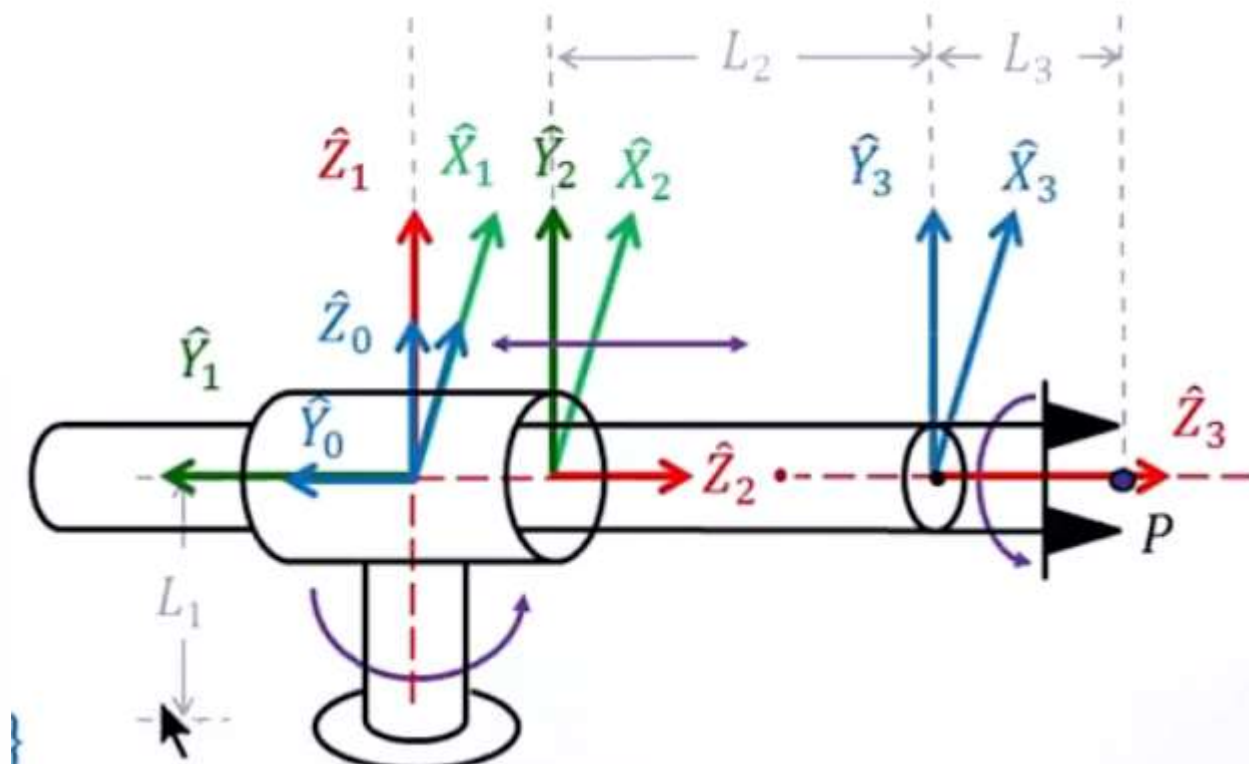
关节轴

公垂线

$\hat{Z}_i$

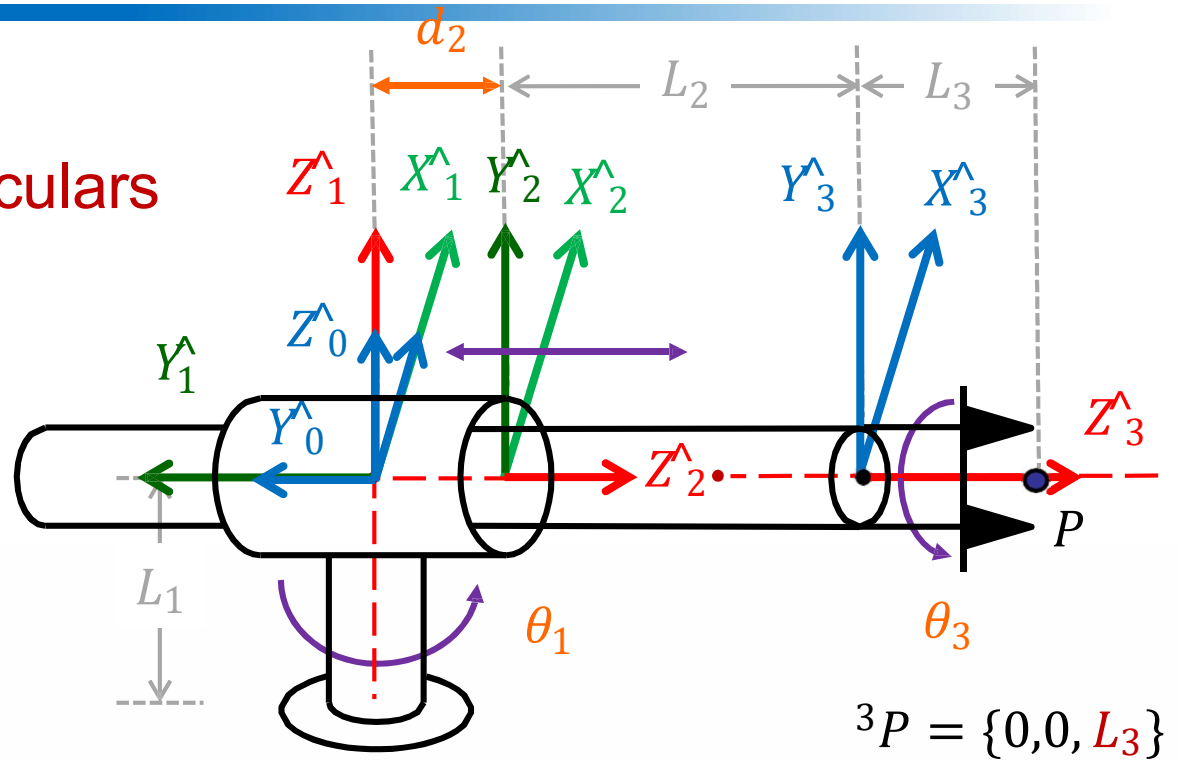
$\hat{X}_i$

$\hat{Y}_i$



Example: A RPR Manipulator -2

- 关节 轴
- Common perpendiculars
- Z_i
- X_i
- Y_i
- 坐标系 $\{0\}$ and $\{n\}$



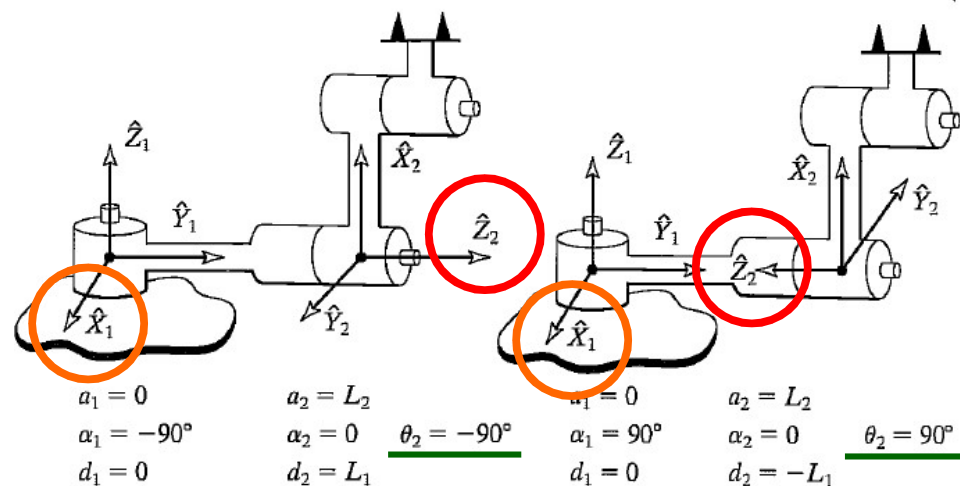
i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	90°	0	d_2	0
3	0	0	L_2	θ_3

Example: A RRR Manipulator -3

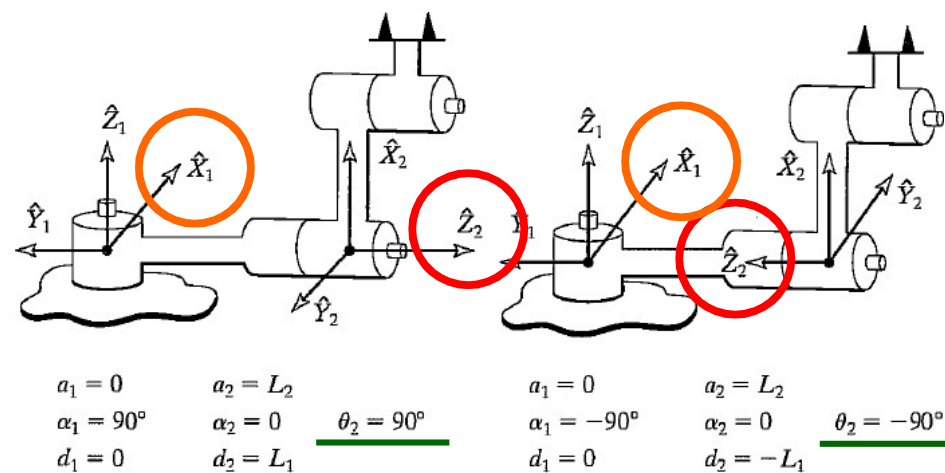
□ 当 $a_1 = 0$

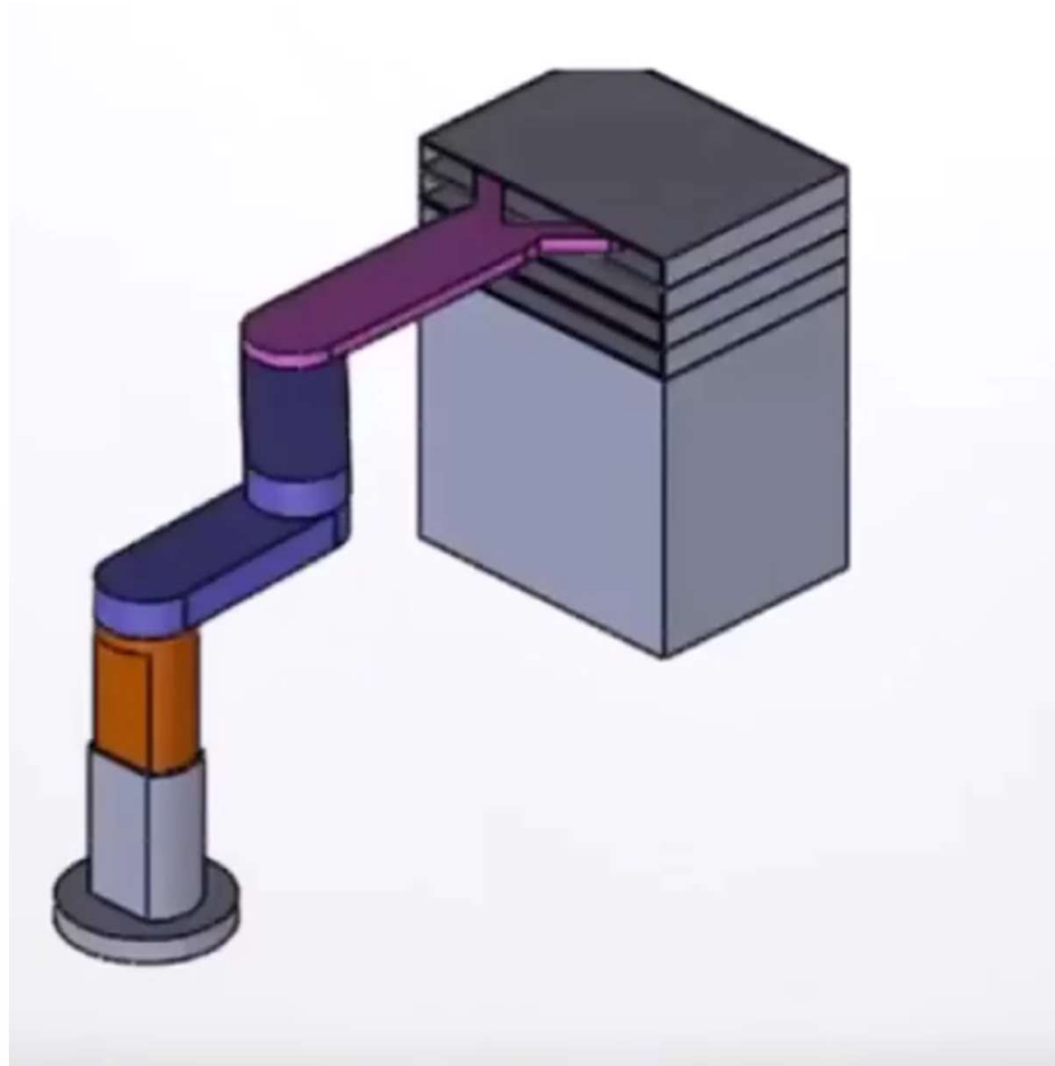
Z_1 和 Z_2 相交

◆ Z_2 两个选择



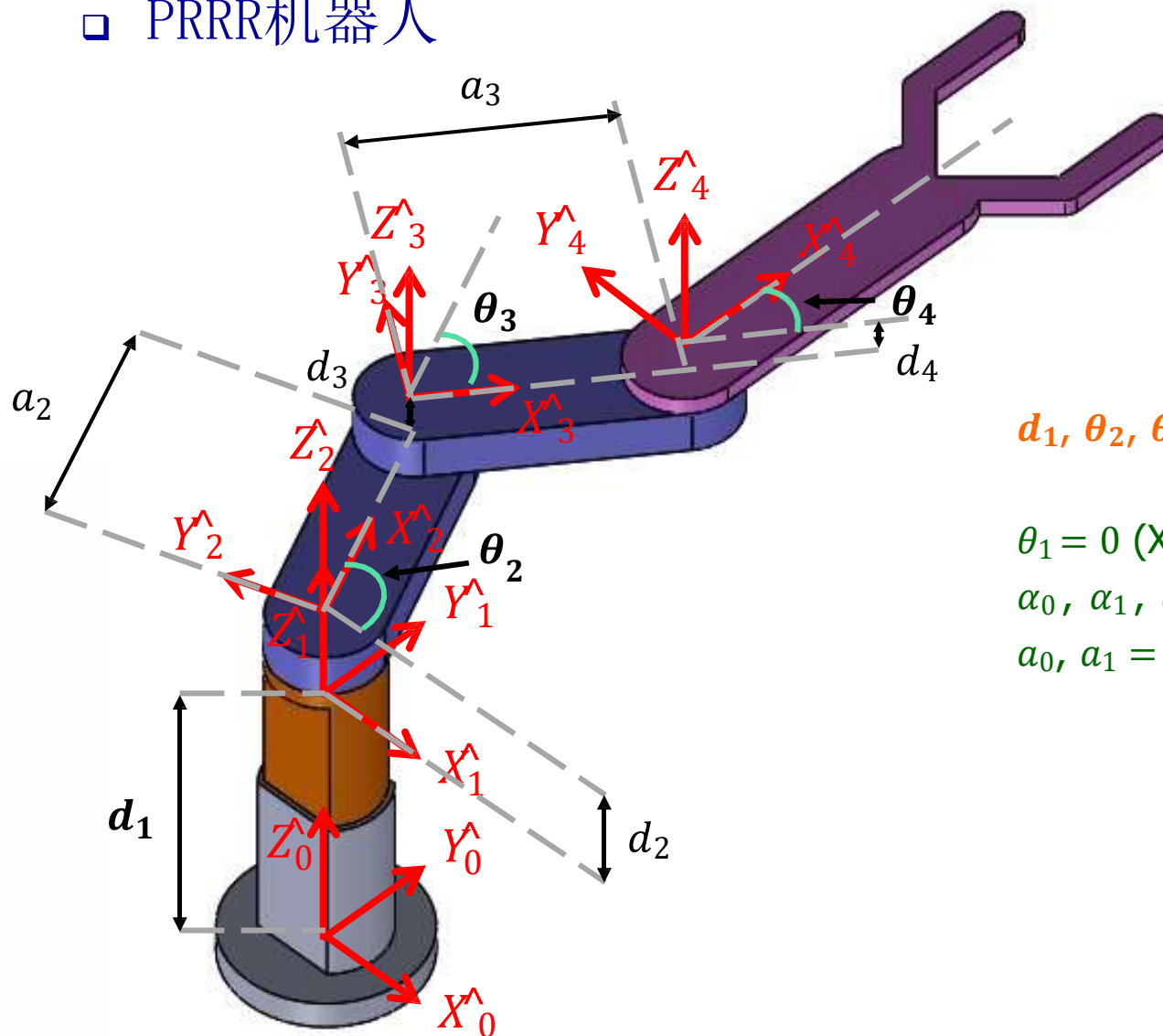
◆ X_1 两个选择





Example: A PRRR Manipulator

□ PRRR机器人



$d_1, \theta_2, \theta_3, \theta_4$ 为驱动关节

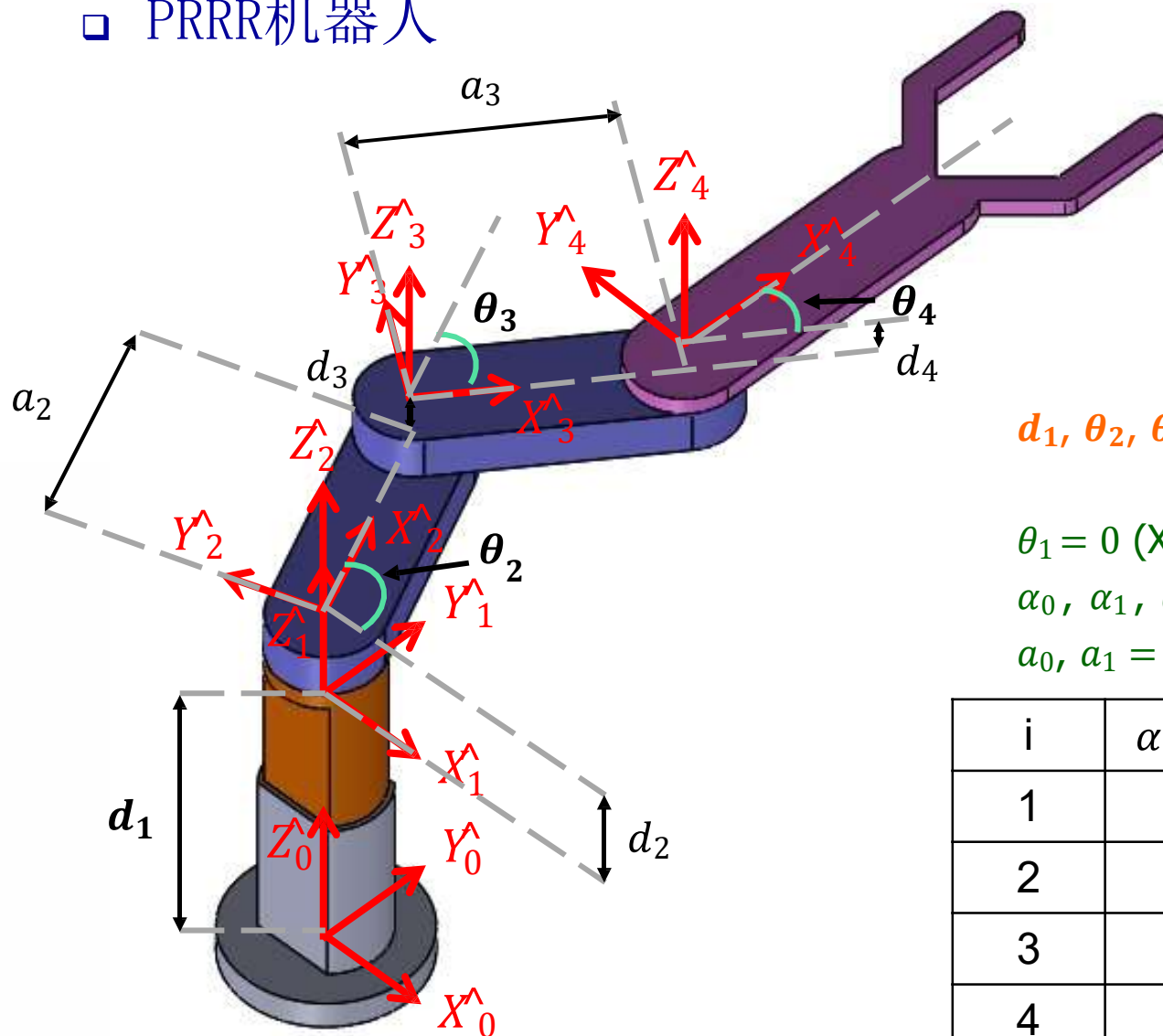
$\theta_1 = 0$ ($X_0 \parallel X_1$)

$\alpha_0, \alpha_1, \alpha_2, \alpha_3 = 0$ (所有Z轴相互平行)

$a_0, a_1 = 0$

Example: A PRRR Manipulator

□ PRRR机器人



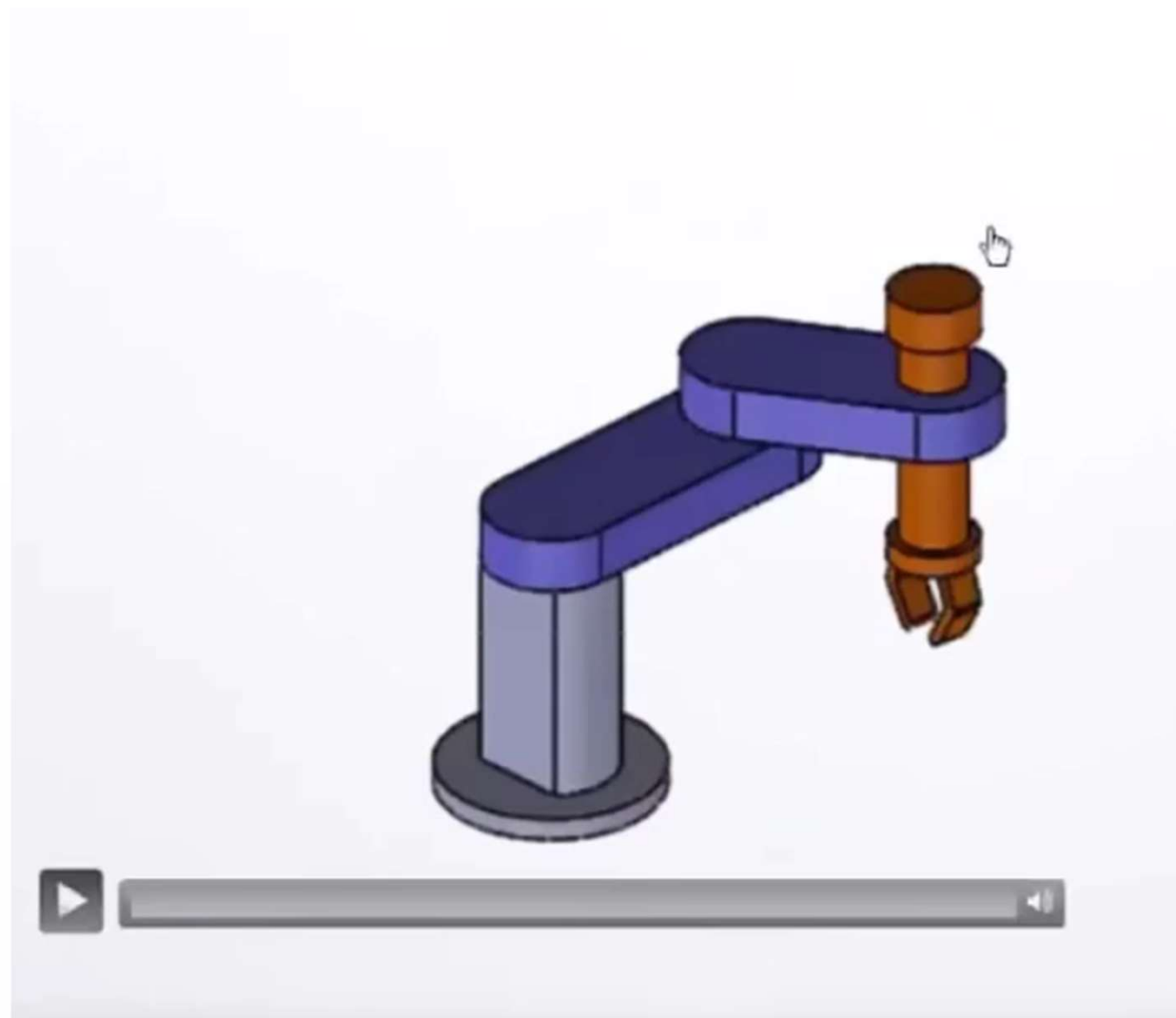
$d_1, \theta_2, \theta_3, \theta_4$ 为驱动关节

$\theta_1 = 0$ ($X_0 \parallel X_1$)

$\alpha_0, \alpha_1, \alpha_2, \alpha_3 = 0$ (所有Z轴相互平行)

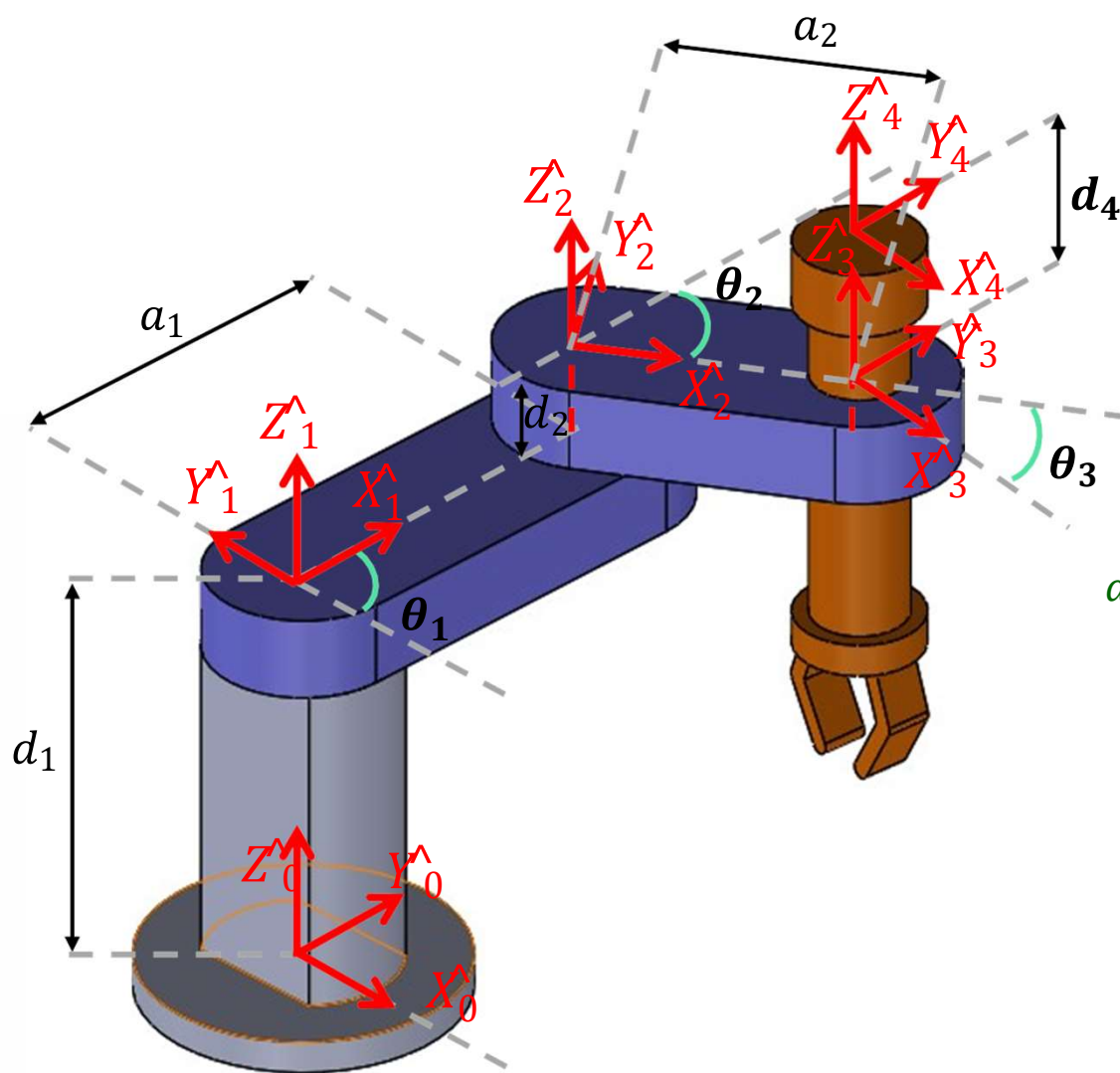
$a_0, a_1 = 0$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	d_1	0
2	0	0	d_2	θ_2
3	0	a_2	d_3	θ_3
4	0	a_3	d_4	θ_4



Example: A RRRP Manipulator

□ SCARA机器人



$\theta_1, \theta_2, \theta_3, d_4$ 为驱动关节

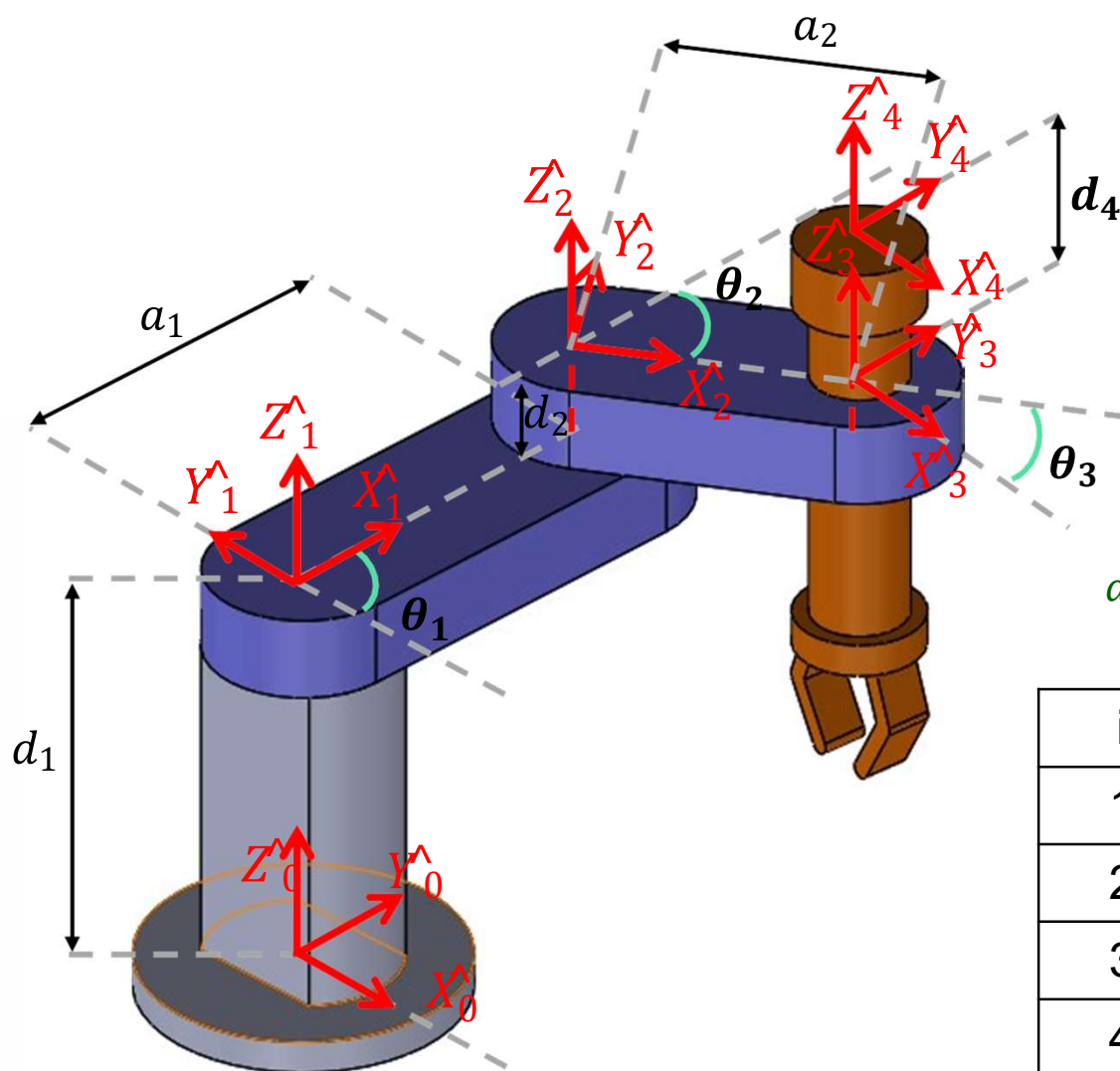
$\theta_4 = 0$ ($X_0 \parallel X_1$)

$\alpha_0, \alpha_1, \alpha_2, \alpha_3 = 0$ (所有Z轴相互平行)

$a_0, a_3 = 0$

Example: A RRRP Manipulator

SCARA机器人



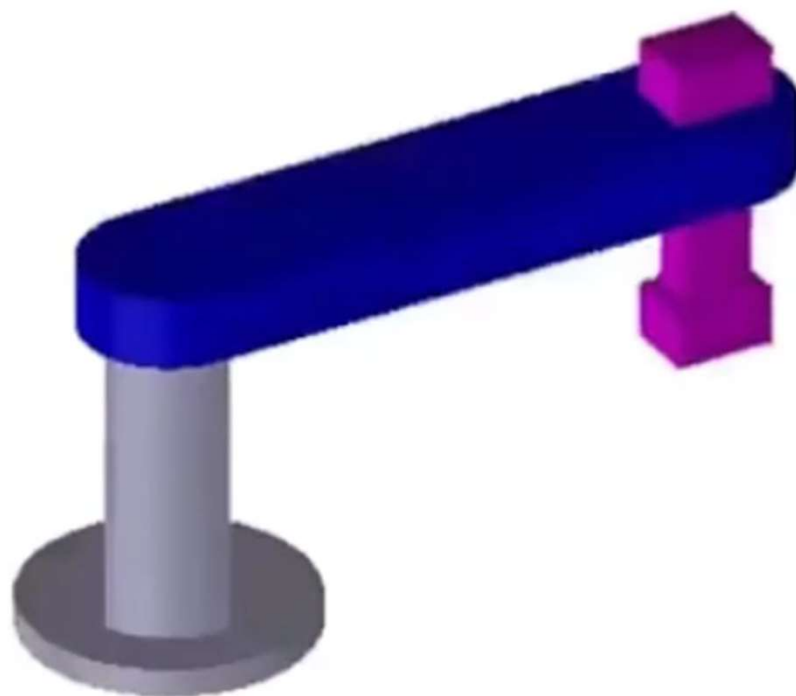
$\theta_1, \theta_2, \theta_3, d_4$ 为驱动关节

$\theta_4 = 0$ ($X_0 \parallel X_1$)

$\alpha_0, \alpha_1, \alpha_2, \alpha_3 = 0$ (所有Z轴相互平行)

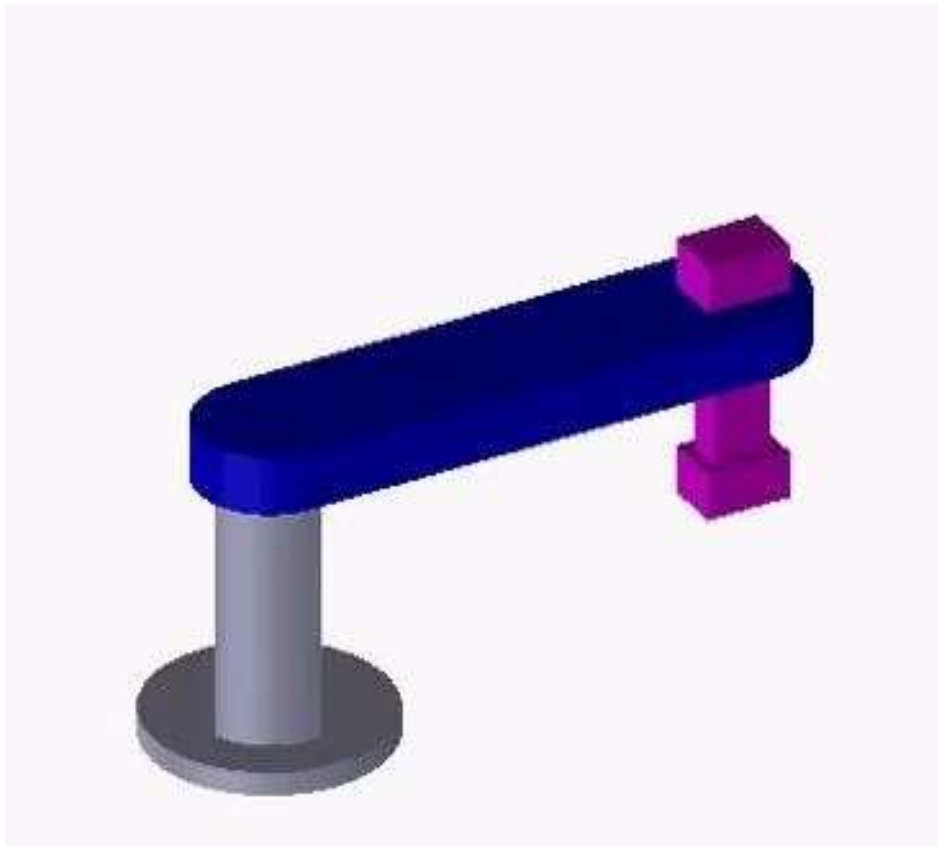
$a_0, a_3 = 0$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	d_1	θ_1
2	0	a_1	d_2	θ_2
3	0	a_2	0	θ_3
4	0	0	d_4	0



Example: A RP manipulator

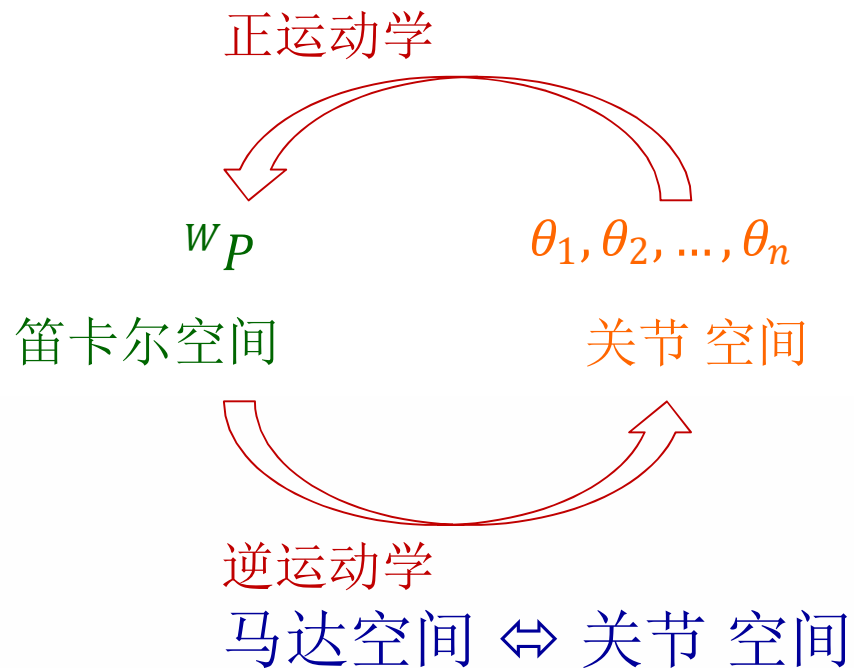
- In-video Quiz: 下方手臂由一个旋转关节和一个移动关节组成，在所有的DH参数 (a_{i-1} α_{i-1} d_i θ_i) 中，哪两个参数为驱动关节？



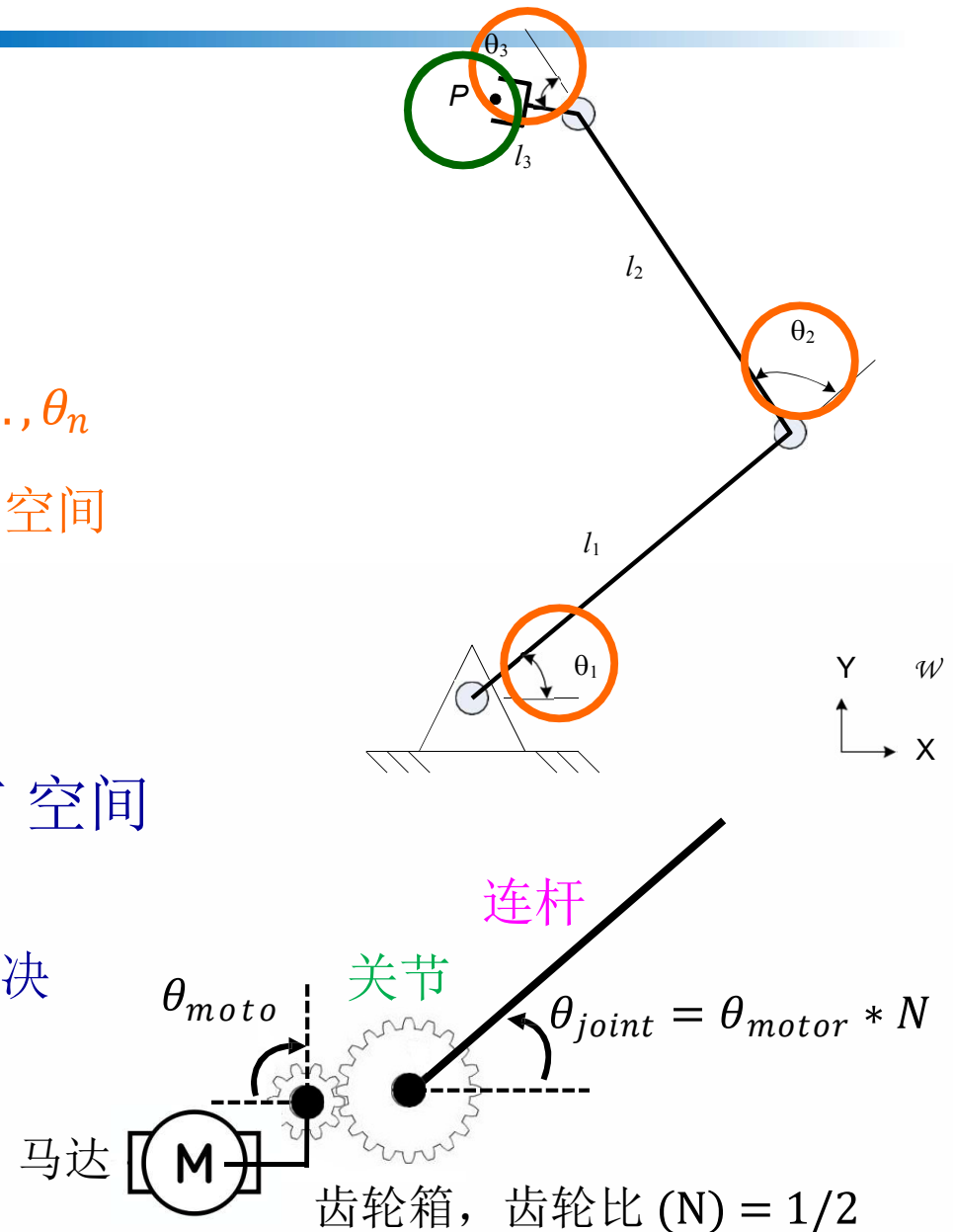
- (A) α_1, a_2
- (B) α_1, d_2
- (C) θ_1, a_2
- (D) θ_1, d_2

Actuator, 关节, and Cartesian Spaces -1

□ 关节空间 $\Leftrightarrow$ 笛卡尔空间

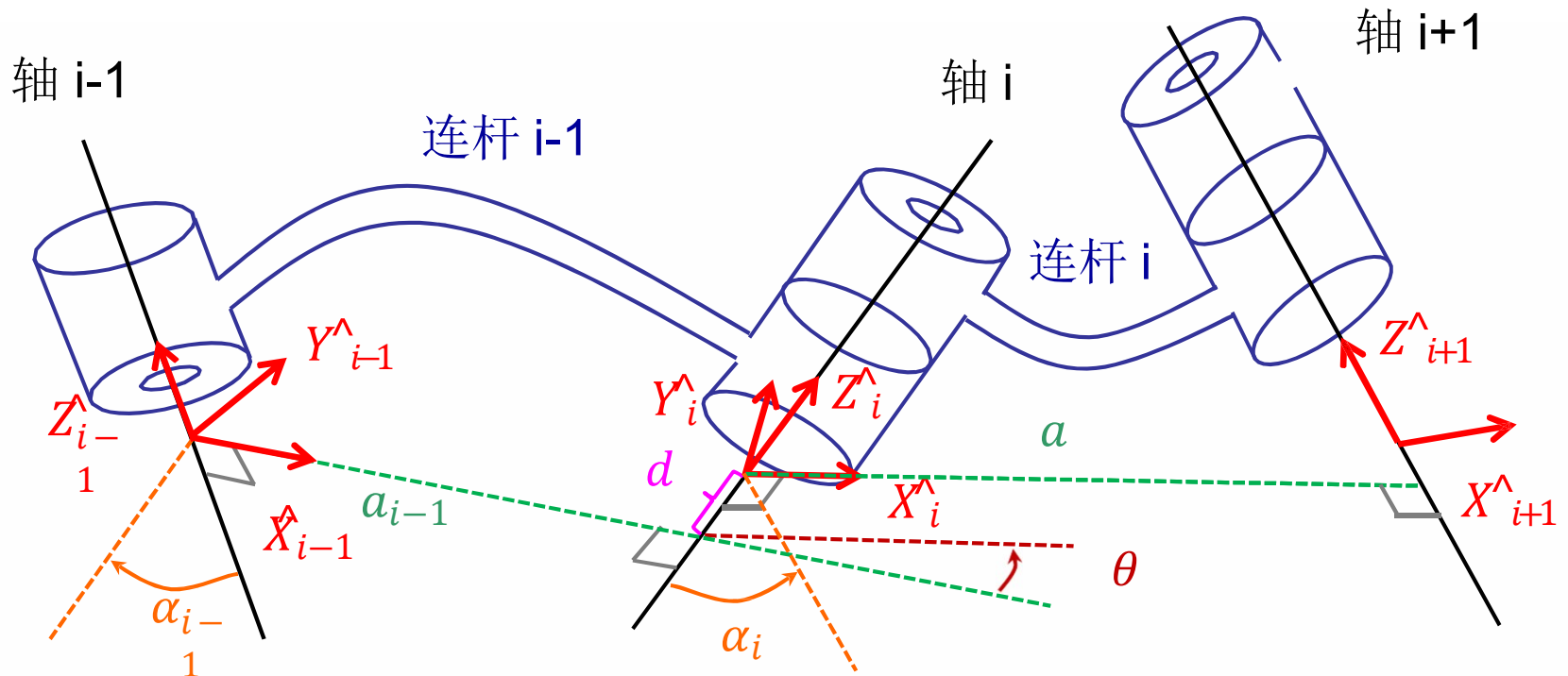


- ◆ 由连结致动器和关节的机构决定



Denavit-Hartenberg表达法小结 (Craig version) -1

- α_{i-1} : 以 $\hat{X}_{i-1}$ 方向看, $\hat{Z}_{i-1}$ 和 $\hat{Z}_i$ 间的夹角
- a_{i-1} : 沿着 $\hat{X}_{i-1}$ 方向, $\hat{Z}_{i-1}$ 和 $\hat{Z}_i$ 间的距离 ($a_i > 0$)
- θ_i : 以 $\hat{Z}_i$ 方向看, $\hat{X}_{i-1}$ 和 $\hat{X}_i$ 间的夹角
- d_i : 沿着 $\hat{Z}_i$ 方向, $\hat{X}_{i-1}$ 和 $\hat{X}_i$ 间的距离

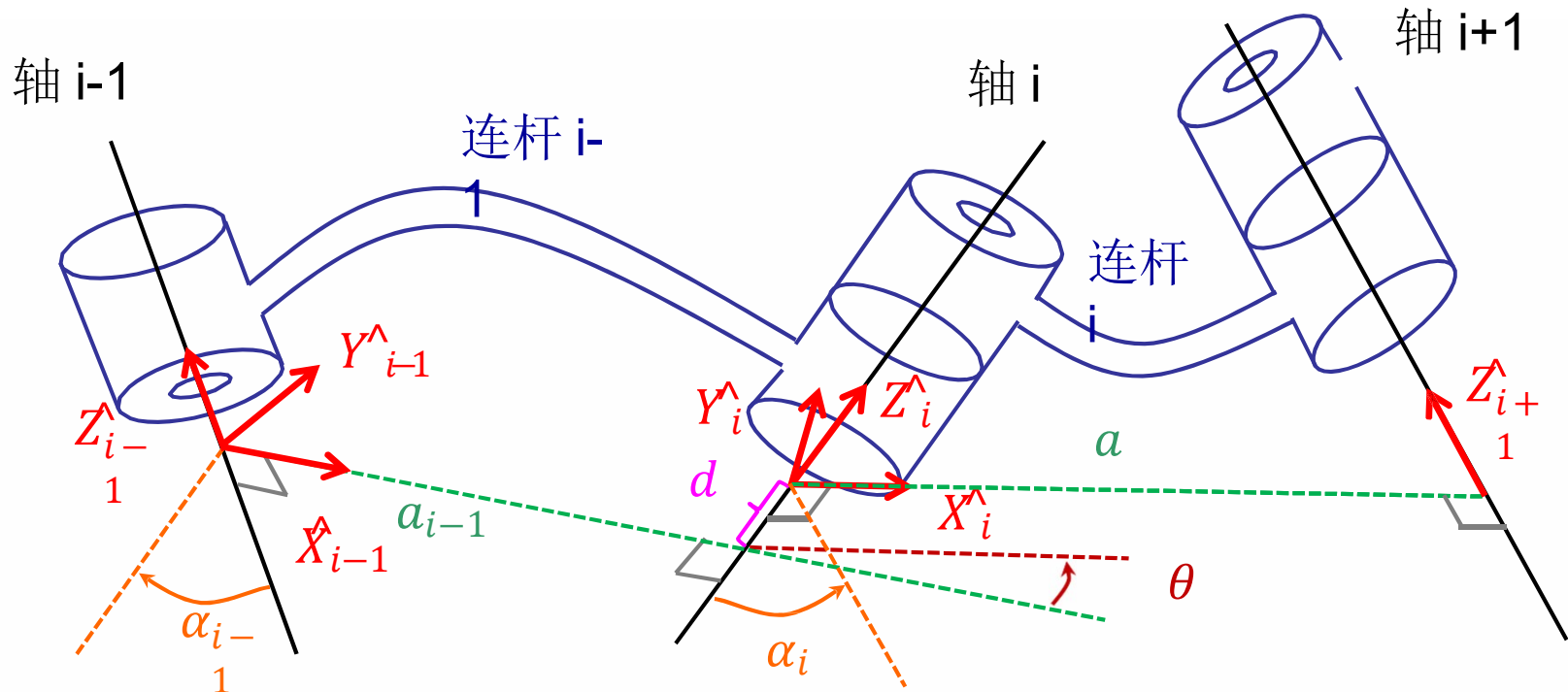


Denavit-Hartenberg表达法小结 (Craig version) -2

$${}^{i-1}T = {}^{i-1}T_R^R T_Q^Q T_P^P T_i^i$$

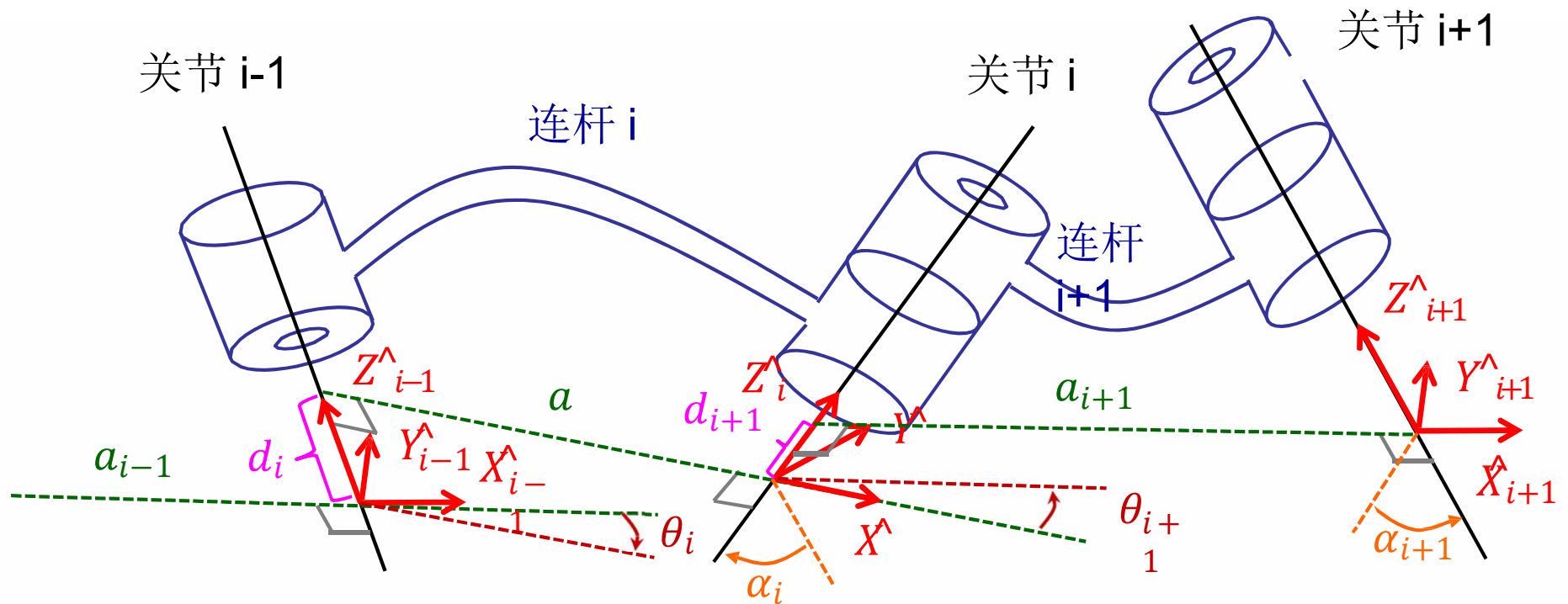
$$= T_{X_{i-1}}(\alpha_{i-1}) T_{X_R}(a_{i-1}) T_{Z_Q}(\theta_i) T_{Z_P}(d_i)$$

$$= \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



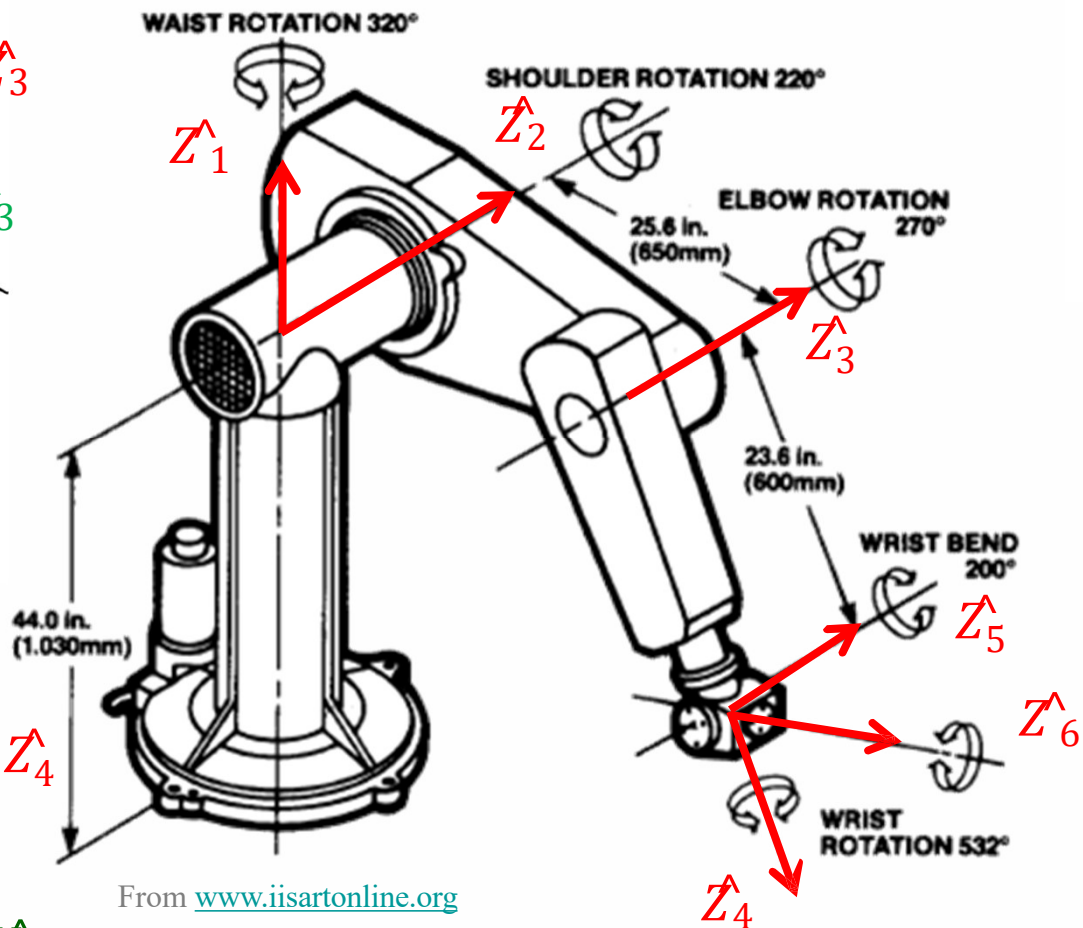
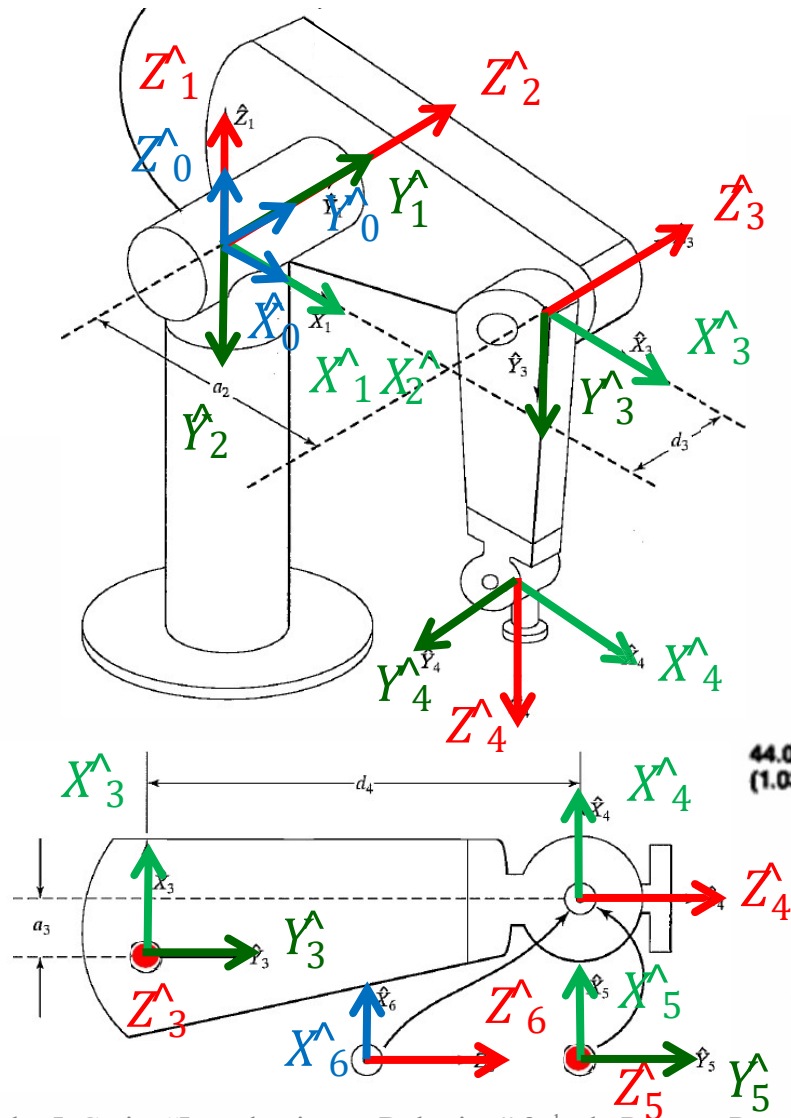
Denavit-Hartenberg表达法 (Standard) -1

- θ_i : 以 Z_{i-1} 方向看, X_{i-1} 和 X_i 间的夹角
- d_i : 沿着 Z_{i-1} 方向, X_{i-1} 和 X_i 间的距离
- a_i : 沿着 X_i 方向, Z_{i-1} 和 Z_i 间的距离 ($a_i > 0$)
- α_i : 以 X_i 方向看, Z_{i-1} 和 Z_i 间的夹角



Example: PUMA 560 -1

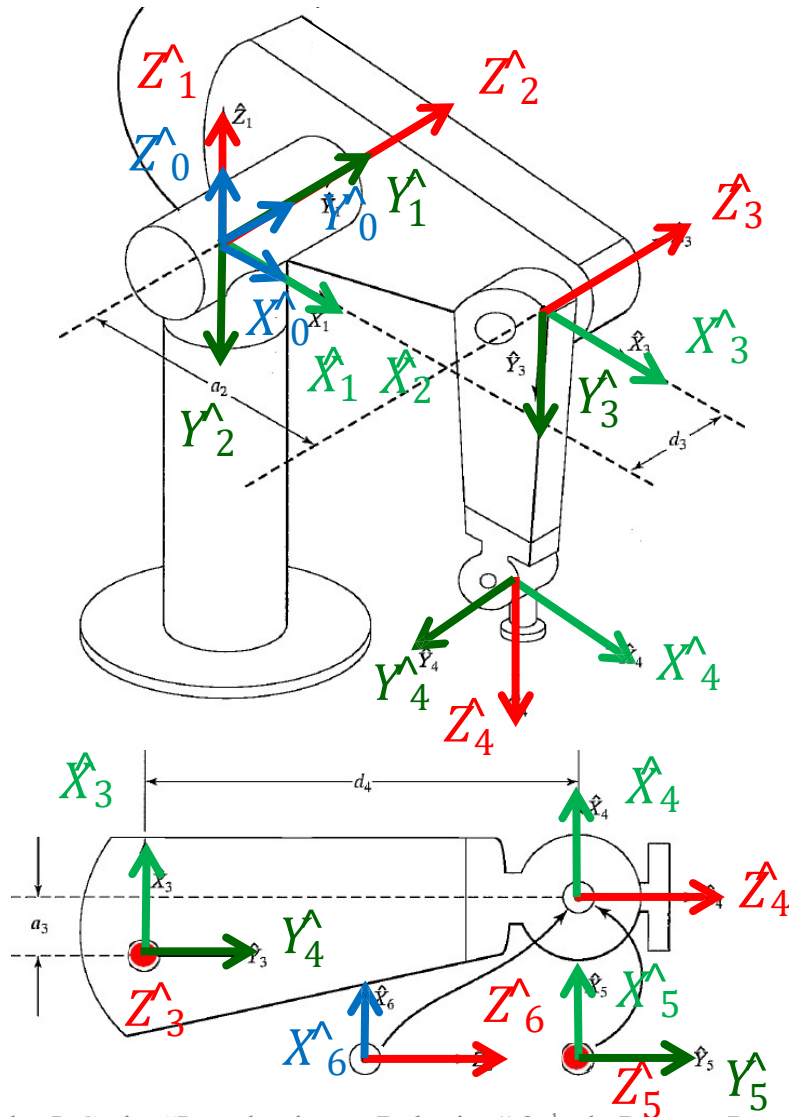
□ 坐标系 (Craig)



From www.iisartonline.org

Example: PUMA 560 -2

□ DH parameters (Craig)



i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0°	0	0	θ_1
2	-90°	0	0	θ_2
3	0°	a_2	d_3	θ_3
4	-90°	a_3	d_4	θ_4
5	90°	0	0	θ_5
6	-90°	0	0	θ_6

Example: PUMA 560 -3

□ Transformation matrices

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4T = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & a_3 \\ 0 & 0 & 1 & d_4 \\ -s\theta_4 & -c\theta_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_2 & -c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^4_5T = \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_5 & c\theta_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^5_6T = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_6 & -c\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example: PUMA 560 -4

□ Combining transformation matrices -1

$${}^4T_6 = {}^4T_5 {}^5T_6 = \begin{bmatrix} c_5 c_6 & -c_5 s_6 & -s_5 & 0 \\ s_6 & c_6 & 0 & 0 \\ s_5 c_6 & -s_5 s_6 & c_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$${}^3T_6 = {}^3T_4 {}^4T_6 = \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & -c_4 s_5 & a_3 \\ s_5 c_6 & -s_5 s_6 & c_5 & d_4 \\ -s_4 c_5 c_6 - c_4 s_6 & s_4 c_5 s_6 - c_4 c_6 & s_4 s_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1T_3 = {}^1T_2 {}^2T_3 = \begin{bmatrix} c_{23} & -s_{23} & 0 & a_2 c_2 \\ 0 & 0 & 1 & d_3 \\ -s_{23} & -c_{23} & 0 & -a_2 s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example: PUMA 560 -5

□ Combining transformation matrices -2

$${}^1_6T = {}^1_3T {}^3_6T = \begin{bmatrix} {}^1r_{11} & {}^1r_{12} & {}^1r_{13} & {}^1p_x \\ {}^1r_{21} & {}^1r_{22} & {}^1r_{23} & {}^1p_y \\ {}^1r_{31} & {}^1r_{32} & {}^1r_{33} & {}^1p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1r_{11} = c_{23}[c_4c_5c_6 - s_4s_6] - s_{23}s_5s_6$$

$${}^1r_{21} = -s_4c_5c_6 - c_4s_6$$

$${}^1r_{31} = -s_{23}[c_4c_5c_6 - s_4s_6] - c_{23}s_5c_6$$

$${}^1r_{12} = -c_{23}[c_4c_5s_6 + s_4c_6] + s_{23}s_5s_6$$

$${}^1r_{22} = s_4c_5s_6 - c_4c_6$$

$${}^1r_{32} = s_{23}[c_4c_5s_6 + s_4c_6] + c_{23}s_5s_6$$

$${}^1r_{13} = -c_{23}c_4s_5 - s_{23}c_5$$

$${}^1r_{23} = s_4s_5$$

$${}^1r_{33} = s_{23}c_4s_5 - c_{23}c_5$$

$${}^1p_x = a_2c_2 + a_3c_{23} - d_4s_{23}$$

$${}^1p_y = d_3$$

$${}^1p_z = -a_3s_{23} - a_2s_2 - d_4c_{23}$$

Example: PUMA 560 -6

□ Combining transformation matrices -3

$${}^0T_6 = {}^0T_1 {}^1T_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$r_{11} = c_1[c_{23}(c_4c_5c_6 - s_4s_5) - s_{23}s_5c_5] + s_1(s_4c_5c_6 + c_4s_6)$$

$$r_{21} = s_1[c_{23}(c_4c_5c_6 - s_4s_6) - s_{23}s_5c_6] - c_1(s_4c_5c_6 + c_4s_6)$$

$$r_{31} = -s_{23}(c_4c_5c_6 - s_4s_6) - c_{23}s_5c_6$$

$$r_{12} = c_1[c_{23}(-c_4c_5s_6 - s_4c_6) + s_{23}s_5s_6] + s_1(c_4c_6 - s_4c_5s_6)$$

$$r_{22} = s_1[s_{23}(-c_4c_5s_6 - s_4c_6) + s_{23}s_5s_6] - c_1(c_4c_6 - s_4c_5s_6)$$

$$r_{32} = -s_{23}(-c_4c_5s_6 - s_4c_6) + c_{23}s_5s_6$$

$$r_{13} = -c_1(c_{23}c_4s_5 + s_{23}c_5) - s_1s_4s_5$$

$$r_{23} = -s_1(c_{23}c_4s_5 + s_{23}c_5) + c_1s_4s_5$$

$$r_{33} = s_{23}c_4s_5 - c_{23}c_5$$

$$p_x = c_1[a_2c_2 + a_3c_{23} - d_4s_{23}] - d_3s_1$$

$$p_y = s_1[a_2c_2 + a_3c_{23} - d_4s_{23}] + d_3c_1$$

$$p_z = -a_3s_{23} - a_2s_2 - d_4c_{23}$$